The Intensive Margin of Small-Firm Protection: Bidding, Rents, and the Fiscal Cost of Public Procurement Set-Asides*

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Abstract

Small-firm procurement preferences are usually priced as an entry problem: fewer bidders, higher prices. This paper shows the larger channel operates inside the auction, not at its door. Exploiting a discrete March 2018 legal reinterpretation that extended São Paulo's SME-only regime to medical supplies—the last exempt product group—I estimate a staggered difference-in-differences against 76 never-treated controls. Winning prices rose 10–11% (7–10% real), narrowing the gap to the buyer's reference ceiling by six percentage points. A Gelbach (2016) decomposition leaves the effect essentially intact: entry and winner-composition channels offset one another, and the within-auction coefficient is if anything larger after conditioning on bidder number and SME identity. Phase-2 bid-moment evidence corroborates—the winning-bid discount widens 5.8 pp, a shift robust to holding firm count fixed. Because 92% of procurement value sits in the top quartile, a value-threshold exemption recovers 90% of the R$50–74 million (US$14–21 million) cost while preserving the preference on 75% of items. The distortion doubles in price-regulated pharmaceuticals—two regulatory regimes interacting perversely. Set-asides do not empty the auction room; they soften every bid inside it.

Keywords: public procurement, SMEs, policy costs, restricted public tenders.

* *This version: April 2026 (First version: January 2021).*

An additional bidder is worth more to the auctioneer than any sophisticated bidding mechanism. — Bulow and Klemperer (1996), paraphrased.

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JEL No.: H32, H57, L26, L53, O12, R12.

1. Introduction

When a government forces its public buyers to procure from small firms only, do prices rise because fewer firms bid, or because each firm bids less aggressively? Four decades of research on bid preferences and set-asides—from Marion (2007) through Krasnokutskaya and Seim (2011), Athey et al. (2013), and Nakabayashi (2013), covering billions of dollars in policy evaluation—have not answered the question directly. The reason is data: existing strategies back out the bidding channel from cross-sectional variation in winning prices alone, a design that conflates entry with aggressiveness by construction. The bid distribution itself has remained a black box.

The answer matters. Set-asides for small and medium-sized enterprises (SMEs) are the most widespread procurement-preference instrument worldwide: the European Union channels a large share of public contract value to SMEs under similar arrangements (PwC, 2014; Loader, 2013, 2015), OECD and SIGMA surveys document comparable regimes across more than one hundred jurisdictions (OECD, 2018; SIGMA, 2016; Hoekman and Tas, 2020), and Brazilian federal law mandates exclusive SME tenders for items valued below R$80,000 (US$22,900),[1] covering approximately 97% of firms representing half of formal employment (Bastos et al., 2018; Thai, 2017; Bosio et al., 2022). A substantial literature shows that these policies boost firm growth, employment, and supplier-network expansion (Mel et al., 2008; Freedman, 2013; Szerman, 2023; Cardoza et al., 2016; Fafchamps et al., 2014; McKenzie and Woodruff, 2015; Ferraz et al., 2016; Cabral, 2017). But the design of the fix depends on the channel. If the price premium runs through the extensive margin, expanding the SME roster—through subsidies, registration

[1] All USD figures in this paper use a reference exchange rate of R$3.50 per US dollar, the average of the BCB monthly closing rate over the September 2016–August 2019 sample period. The 2016–2019 annual averages were R$3.49 (2016), R$3.19 (2017), R$3.65 (2018) and R$3.88 (2019).

relief, or capacity-building—can lower the fiscal cost without weakening the preference. If instead it runs through the intensive margin, no roster expansion will help: the only lever is to shrink the *scope* of the rule itself. Policy design hinges on the answer—for the procurement officers in Brazil's 5,570 municipalities implementing Law 14,133/2021, for the 130 countries with SME-preference regimes in place, and for development economists wrestling with the trade-off between competitive efficiency and distributional objectives in public spending.

Theory points both ways. Bulow and Klemperer (1996) show that, across a wide class of auction environments, an additional bidder is worth more to the auctioneer than any sophisticated mechanism—a headline result for the extensive margin. Krasnokutskaya and Seim (2011) formalize bid preferences in the procurement context and show that both margins are operative. Which dominates quantitatively is an empirical question, and outside the US the evidence for SME set-asides is essentially absent (Nakabayashi, 2013; Szucs, 2024; Decarolis et al., 2025).

This paper shows that the price effect is not a pure headcount effect and that an important within-auction bidding channel—the *intensive margin*, how each participating firm bids—is the most plausible reading of what else is going on. Think of each auction as a bidding classroom: the SME-only rule does not merely empty some desks; it changes how each remaining student answers. Using phase-2 electronic-auction bid-distribution moments from 3.7 million procurement observations on Sao Paulo's BEC platform, I show that when Brazil's March 2018 reinterpretation extended the state's SME-only regime to medical supplies—the only previously exempt product group—the winning bid immediately moved about six percentage points closer to the buyer's reference price, an effect that persists after conditioning on exactly two, three, or five-plus firms. A Gelbach (2016) decomposition finds that entry and winner-composition channels partially offset one another: individually large but opposite-signed, they net to roughly five percent

of the 10–11% price increase, leaving approximately ninety-five percent unexplained by these observable channels. Phase-2 bid moments—direct measurements of within-auction bidder behavior, not residuals from a mediation exercise—then document that the bid distribution itself changes when the restriction binds, pointing to the intensive margin as the most plausible interpretation of the residual. Pairing a discrete policy cutoff with within-auction bid-distribution data sidesteps the identification problems that have constrained cross-sectional studies of bid preferences (Marion, 2007; Krasnokutskaya and Seim, 2011; Athey et al., 2013).

The quasi-experiment I exploit has three features that sharpen identification. First, the SME set-aside in Sao Paulo operates through *opt-out costs*: public buyer units (PBUs) that wish to run open tenders must justify the departure in a formal report to the audit courts, which is costly, scrutinized, and sanctionable. The rule effectively penalizes the efficient procurement choice, shifting the default from open to restricted tendering. Second, between August 2014 and February 2018, Group 65 (medical and hospital supplies) was singled out as the sole product group exempted from the state's SME-only rule, under a joint agreement between the state government and the audit court (TCE-SP) that medical supplies warranted open competition. Third, in March 2018 a legal opinion from a separate oversight agency (PGE-SP) reversed the exemption on isonomy grounds, extending opt-out costs to Group 65 along with every other product group. Because the reversal was driven by a legalistic reinterpretation rather than by any change in procurement outcomes for Group 65, the switch is plausibly exogenous.

The March-2018 event is, in modern language, a single-cohort staggered difference-in-differences (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021; Borusyak et al., 2024): Group 65 is the lone treated cohort, and the 76 other product groups—subject to the SME-only rule since the 2014 federal mandate—are never-treated controls within the sample window. Identification therefore comes from a clean treated-

vs-never-treated comparison, free of the "forbidden" implicit comparisons that motivate the modern staggered-DiD literature. The Goodman-Bacon decomposition confirms the structure, and the Kim and Lee (2019) difference-in-differences-in-reverse framework reformulates the same estimand under the complementary view that the 76 controls are "always-treated." Both framings deliver the same coefficient and are numerically equivalent to the Sun-Abraham interaction-weighted estimator for a single-cohort design.

Imposing the SME-only rule on Group 65 raises procurement prices by 10–11% in the 18-month window (13–14% in the 6-month window, 7–10% in IPCA-deflated real terms), reduces participating firms by 10–11%, and cuts valid bids by 7–8%. Winning firms under open tenders sit 6–14 km further from buyers, so the SME preference does deliver geographic proximity even as it raises fiscal cost. The mechanism shows up directly in the phase-2 bid moments: under open tenders the winning bid reaches about six percentage points deeper below the reference price, an effect that survives conditioning on exactly two, three, or five-plus firms. These bid-level moments are direct measurements of within-auction bidder behavior. Under the two-way fixed-effects specification, the bid-SD evidence is less clean than the bid-discount evidence (the sample-wide SD coefficient does not clearly tighten with firm count), so the paper rests its intensive-margin interpretation on the discount shift and the Gelbach residual rather than on SD compression. The Gelbach (2016) decomposition corroborates the interpretation: entry and winner-composition channels partially offset, with a net contribution of about five percent to the price effect; approximately ninety-five percent is unexplained by those channels, consistent with a within-auction bidding-behavior story.

Identification varies by outcome and is worth stating honestly. The log-price effect is the paper's core causal estimate: the price placebo is small in magnitude relative to the headline (between one-quarter and one-third), though two of three placebo cutoffs are marginally significant under the two-way FE specification. The log firms and log bids

coefficients, by contrast, have pre-trend contamination that neither placebos nor Sun-Abraham diagnostics fully resolve—they are reported as suggestive supporting evidence, not as independent causal estimates. The distance outcome is causally identified (placebos are null); and the bid-distribution moments that drive the mechanism section are within-item and independent of the firms-bids pre-trend issue. Three estimators from the staggered-DiD literature (CS2021, Sun-Abraham, Goodman-Bacon) deliver the same sign on all four outcomes; the item-level Sun-Abraham estimate is numerically equivalent to the headline DiD, while the group-month aggregations from CS2021 and Goodman-Bacon differ in magnitude as expected. One hundred percent of the Goodman-Bacon weight falls on the clean treated-vs-never-treated comparison: no "forbidden" implicit comparisons enter. A formal spillover test on the 76 control groups bounds the implied bias from non-SME firms redirecting into control markets at less than 1% of the headline, preserving the lower-bound interpretation. Linking winners to RAIS (Brazil's matched employer-employee registry) leaves the coefficient invariant to firm-size, age, and two-digit CNAE controls, and the shift in winner composition is recoverable directly from a RAIS-validated SME flag. One caveat is structural: with a single switching unit, classical permutation inference has low power, and the randomization p-value is 0.342. The evidence for causality therefore rests on the consistency of effects across four outcomes and the convergence of multiple estimators rather than on that permutation. Placebo tests, alternative clustering, wild cluster bootstrap, winsorization, Lee bounds, quantile DiD, synthetic control, and causal forest analysis provide further corroboration in the Appendix and Online Appendix.

The static fiscal cost of the SME-only rule for Group 65 alone is R$50–74 million (US$14–21 million) over the 18-month window (R$50 million real, R$74 million nominal),

or 7–11% of the group's procurement value.[2] The number is mechanical—a reduced-form price coefficient times pre-period procurement value—and ignores the completion, entry, and reference-price responses that a structural counterfactual would price. Two findings sharpen the policy implications. First, 92% of that procurement value is concentrated in the top value quartile, so a mechanical value-threshold exemption recovers 90% of the accounting cost while preserving the SME preference on 75% of items by count. SMEs lose access to the top-quartile contracts under this reform—so it is not strictly Pareto-improving, and a full welfare analysis would require modeling the behavioral responses—but even in static terms the reform substantially relaxes the efficiency-equity trade-off that motivates SME-only rules. Second, CMED-regulated pharmaceuticals (class 6531 within Group 65) bear more than twice the distortion of the non-pharma medical supplies: the CMED list-price cap does not shield procurement from the ME/EPP preference, consistent with contract prices lying below the cap and the restriction compressing competition within the residual slack. Two Brazilian regulatory regimes—SME procurement preferences and drug price controls—therefore interact perversely, raising public-health spending above what either instrument alone would deliver.

The paper makes four contributions to the literatures on set-asides, bid preferences, and public procurement efficiency.

(i) Direct bid-distribution evidence on the intensive margin. Prior work on SME preferences has largely inferred the mechanism from changes in the winning price or the number of bidders (Marion, 2007; Athey et al., 2013; Krasnokutskaya and Seim, 2011; Nakabayashi, 2013). Using phase-2 bid moments—the minimum, maximum, mean, and standard deviation of the full bid distribution within each electronic auction—I run two

[2] Three fiscal accounting variants appear in the paper and all bracket the same object under different aggregations: R$50–74 M is the headline range (pooled 18-month coefficient applied to R$689 M pre-period procurement value, real and nominal); R$59 M uses quartile-specific coefficients $\hat{\beta}_q$ applied to within-quartile procurement value (Table 20); R$70.7 M aggregates the pharma vs. non-pharma split specification (Table 21). The R$50–74 M range is the least assumption-heavy.

direct bidder-level tests: (i) the winning bid moves about six percentage points further below the buyer's reference price under open tenders, an effect that survives conditioning on firm count; and (ii) the normalized gap between the runner-up's bid and the winner's bid shrinks by 2.7 percentage points, with the effect concentrated in auctions with five or more firms. The runner-up gap is a direct measurement of competitive clustering at the auction level, not a mediation residual, and its narrowing in auctions where competition has room to bite is difficult to rationalize without a within-auction bidding response. The finding qualifies as what Davis (1971) called a *False Correlation*: entry and bid aggressiveness are theoretically linked but appear quantitatively independent in this setting—95% of the set-aside price effect is unexplained by entry and winner-SME composition (net of two partially offsetting channels), and the direct phase-2 evidence shows that both the winner and the runner-up move when the restriction binds, even with firm count held fixed. This bid-level evidence is consistent with the auction-theoretic prediction of Bulow and Klemperer (1996) and Krasnokutskaya and Seim (2011) and helps reconcile the coexistence of large price effects with modest entry effects that the literature has struggled to explain.

(ii) A sharp counterfactual policy-design result. Because 92% of Group-65 procurement value concentrates in the top quartile, a mechanical value-threshold exemption—applying the SME-only rule only to items below approximately R$7,600 (US$2,180)—recovers 90% of the accounting fiscal cost while preserving the SME preference on 75% of items by count. The exercise is static accounting: it assumes within-quartile price effects are invariant to the reform, which would fail in equilibrium; and it ignores behavioral responses (completion, entry, reference pricing). It nonetheless identifies the natural lever—top-value contracts account for most of the fiscal cost, so threshold-based exemption is the natural design—in a form that the ongoing implementation of Law 14,133/2021, Brazil's new public procurement framework, could use as a starting point rather than a

blueprint.

(iii) A perverse interaction between two Brazilian price-regulation regimes. CMED-regulated pharmaceuticals (class 6531 within Group 65) bear more than twice the price distortion of the non-pharma medical subsegment under the SME-only rule. The list-price cap imposed by CMED—designed to contain public-health expenditures—does not shield procurement from the ME/EPP preference, because actual contract prices sit below the cap and the restriction compresses competition within the residual slack. This interaction is new to the literature on procurement preferences (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2025; Colonnelli and Prem, 2022) and has direct implications for countries that combine SME preferences with pharmaceutical price controls.

(iv) A fiscal-cost estimate from a large developing economy. The paper delivers quasi-experimental estimates of SME set-aside costs in Sao Paulo (R$50–74 million, or US$14–21 million, over 18 months for a single product group), rare in the literature outside the United States and large firms in Europe (Athey et al., 2013; Krasnokutskaya and Seim, 2011; Bandiera et al., 2009; Best et al., 2023). The design—a single switching cohort against 76 never-treated controls—is a *staggered-DiD reversal* rarely exploited in applied work (Kim and Lee, 2019; Athey and Imbens, 2022), and the estimates are robust to the Callaway-Sant'Anna (2021), Sun-Abraham (2021), and Goodman-Bacon (2021) estimators, to a formal cross-group spillover test, and to winner-level controls for firm size, age, and sector extracted from Brazil's matched employer-employee registry (RAIS).

Section 2 provides the institutional background. Section 3 describes the data. Section 4 presents the empirical analysis, heterogeneity, robustness, and fiscal cost quantification. Section 5 concludes with policy implications.

2. Institutional Background

Brazil's federal SME statute (Complementary Law 123/2006, amended 2014) defines micro- and small enterprises by annual gross revenue—R$360,000 (US$103,000) and R$4,800,000 (US$1.37 million) ceilings, respectively—and mandates exclusive SME tenders for items valued at R$80,000 (US$22,900) or less. Public buyer units (PBUs)—federal, state, and municipal administrative entities subject to procurement rules—can avoid the SME-only default only by justifying the departure through a formal report to the audit courts, a costly and scrutinized process. The 2014 amendment converted what had been an option ("may") into a binding default ("must") and imposed these *opt-out costs*; state-level adherence to SME-only tendering for eligible items rose from roughly 13% to nearly 70% in the subsequent years.

Since 2005, all public procurement in the state of Sao Paulo passes through Bolsa Eletrônica de Compras (BEC), a centralized electronic reverse-auction platform in which registered suppliers submit sealed bids (phase 1) followed by an iterative electronic auction (phases 2 onward) in which each participant may progressively lower its offer until the auction closes. BEC handled approximately R$13 billion (US$3.7 billion) of procurement in 2019 alone and has cumulatively moved more than R$105 billion (US$30 billion) since implementation, spanning 860,000 purchase offers and 4.8 million item-level transactions. BEC organizes its catalog into three tiers—group, class, and item—so that "medical, dental, and hospital equipment and supplies" form Group 65, within which class 6531 identifies medications subject to price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED).

Group 65 was a singular exception to the 2014 rule. Between August 2014 and February 2018, the state government and the Tribunal de Contas do Estado (TCE-SP) operated under a joint interpretation that medical supplies were strategic goods requiring open competition; during that window no Group-65 bid was restricted to SMEs. In November

2017, the Procuradoria-Geral do Estado (PGE-SP, a separate control agency) issued a legal opinion reversing the exception on isonomy grounds—the constitutional requirement that comparable cases be treated alike. The opinion argued that no substantive feature of medical supplies justified procedural distinctions from other product groups. Effective March 2018, opt-out costs extended to Group 65; SME-only adherence within the group jumped to 43% on average, below the rate for other groups largely because class-6531 pharmaceuticals are oligopolistic and often fail the three-eligible-SMEs requirement.

What matters for identification is that the policy change was triggered by a legalistic reinterpretation of the isonomy principle at PGE-SP, not by anything happening in the Group-65 procurement market. Public documentation turns up no sign that the decision was a response to price levels, competition patterns, or supplier complaints specific to Group 65 in 2017; the timing was set by the bureaucratic process inside PGE-SP and by when that office chose to communicate the opinion to PBUs. The change is therefore plausibly exogenous to the outcomes studied. This is the institutional basis for the identifying assumption that, absent the March-2018 reinterpretation, Group 65 would have tracked the contemporaneous trajectory of the 76 never-treated product groups.

3. Data

I use bidding-level administrative data on public procurement of common goods and services in the state of Sao Paulo, Brazil, from January 2016 to December 2019. All transactions flow through BEC, the state's centralized electronic procurement platform; SEFAZ/SP (the State Finance Secretariat) manages it and centralizes the bidding records. Because BEC covers the universe of standardized goods procurement in the state, the data are free from the selection concerns that arise with partial or voluntary coverage.

BEC serves 1,344 PBUs as regular buyers: state bureaus from the executive, legislative, and judicial branches of Sao Paulo, along with affiliated entities such as municipal

governments and accredited private organizations. Between 2016 and 2019 these PBUs purchased 82,569 distinct items (goods) across 832,984 successful transactions.

BEC catalogs goods and services at three nested levels: group, class, and item. Health items sit in Group 65 (medical, dental, and hospital equipment and supplies). Within Group 65, class 6531 holds medications with or without ANVISA notification; item 110639, for example, is the drug "Furosemide 40 milligrams, coated tablets, units." To make the policy shift concrete: in February 2018, under the open-tender regime, a Sao Paulo PBU bought 50,000 units of this same furosemide item in an electronic auction where ten firms pushed the bids down to 14 percentage points below the reference price, and the winning unit price cleared about 12% below reference. In October 2018, after the SME-only rule bound Group 65, a comparable furosemide purchase by the same PBU attracted three firms, phase-2 bids clustered within 3 percentage points of the reference, and the winning unit price came in just above reference.[3] The paper quantifies this pattern across the full Group-65 universe.

The data are organized by *purchase offer* (PO)—the electronic document a PBU issues to identify and quantify the goods or services it intends to buy. Each PO has a 22-character code and may list one or more items, but each item runs through its own purchase process. A unique item-level observation is therefore identified by the combination of PO and item codes (POI).

The central regressor, constructed from the item group code, is a binary indicator equal to one if the item belongs to Group 65 and zero otherwise. Group 65 is the *switched group*—the set of items affected by the March-2018 policy change—and accounts for nearly 27% of all purchases between 2016 and 2019. The remaining product groups form the control set.

[3]This stylized pair is illustrative rather than a case study; the average effects across the 101,386 class-6531 observations in the 18-month window are reported in Section 4.3 and the pharma decomposition table.

There are 77 non-Group-65 product groups active over 2016–2019; 76 of them satisfy the pre/post balance requirement of the 18-month analytical window and form the never-treated comparison set for the empirical design. Among those, the most significant are groups 89 (food products), 75 (office supplies), 86 (computer products), and 79 (cleaning materials), representing 37.5% of total purchases in the period. For each POI, there is information about item quantities, bid prices (winners and losers), the number of participant firms, the number of bids, whether the public tender was successful or not, the identification of the PBU, and firms' and PBUs' location, among other variables.

The empirical analysis tracks four outcomes. The *negotiated price* (in logs) is the final transacted price, available only for completed items. The *number of participant firms* (in logs) counts the distinct firms that submitted at least one bid, whether the item was ultimately purchased or not. The *number of valid bids* (in logs) counts the qualifying bids received per item. *Distance* is the kilometer distance from the winning firm's location to the buying PBU. Every regression controls for log quantity ordered and the type of tender (sealed bid vs. reverse auction), and includes item-level fixed effects that absorb all time-invariant heterogeneity across items—differences in markets, pricing norms, and typical supplier pools.

To rule out compositional bias, I restrict each regression to items that appear in both the pre- and post-periods within the relevant time window. Three nested windows (6, 12, and 18 months around the March-2018 cutoff) check that the estimates are stable as the sample expands. Price and distance regressions use the subsample of completed items (those with a winning bid); participation and bid regressions use all items, including those with no winner. Conditioning on completion raises a sample-selection concern—the treatment itself may affect whether an item is transacted—which the Lee bounds analysis in the Appendix addresses directly. Table 1 presents descriptive statistics for the key variables used in the empirical analysis, disaggregated by group (Group 65 vs. Others)

and period (Pre vs. Post the policy shift in March 2018).

Table 1: Descriptive Statistics (18-month window)

	Group 65, Pre		Group 65, Post		Others, Pre		Others, Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Price (levels)	1,402.09	20,563.54	1,507.92	25,872.31	1,465.85	50,385.93	1,291.37	68,042.55
Log price	1.72	2.84	1.91	2.81	2.85	2.12	2.78	2.07
Number of firms	3.17	2.52	3.02	2.40	4.66	3.27	4.69	3.28
Log number of firms	0.88	0.73	0.84	0.71	1.30	0.72	1.30	0.72
Number of valid bids	11.10	15.49	10.99	16.51	10.13	15.74	10.47	16.87
Log number of valid bids	1.64	1.24	1.61	1.23	1.69	1.06	1.70	1.08
Distance (km)	238.54	270.42	215.08	239.63	164.78	182.23	170.28	186.07
N (completed / all)	65,606	77,206	66,822	85,181	246,277	287,500	271,011	323,836

Notes: Price and distance statistics computed on completed items (status = 1). Firm and bid statistics computed on all items.

4. Empirical Strategy and Results

4.1. Main Specification and Identification Strategy

The March 2018 policy change constitutes a single-cohort staggered difference-in-differences design. Group 65 (medical and hospital supplies) enters the SME-only regime in March 2018, joining the 76 other product groups that had been subject to the same restriction since August 2014. Within the 18-month estimation window (September 2016 to August 2019), Group 65 is therefore the sole treated cohort, and the remaining 76 product groups are never-treated controls from the perspective of this specific policy event—they had all transitioned to the SME-only regime before the sample window begins. This structure aligns the design with the modern staggered-DiD literature (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfoeuille, 2020; Roth et al., 2023): identification comes from comparing a single switching cohort to a stable never-treated comparison set, a setting in which the concerns about "forbidden" implicit comparisons that motivate that literature

do not arise.[4]

Section 3 laid out the institutional timeline; Table 2 restates it in treatment-assignment form. From August 2014 through February 2018, every product group except Group 65 was subject to opt-out costs under the SME-only regime; Group 65 operated under mandatory open tenders. From March 2018 onward, Group 65 joined the other 76 groups under the same opt-out costs.

Table 2: Policy status by group and period

Groups	Before March 2018	After March 2018
Group 65 (treated cohort)	Open tenders	SME-only (opt-out costs > 0)
Other 76 groups (never-treated)	SME-only (opt-out costs > 0)	SME-only (opt-out costs > 0)

Using subindex p for purchase offer, i for items, t for months, and g for groups, the estimating equation is:

$$y_{pigt} = \eta_i + \lambda_t + \beta \left(g65_{pgt} \times \mathrm{Pre}_t \right) + x_{pigt}\, \delta + \varepsilon_{pigt} \tag{1}$$

where y is an outcome, η_i is item fixed effects, λ_t is month fixed effects, Pre_t is a dummy equal to one before March 2018, $g65_{pgt}$ is a dummy for Group 65, and x_{pigt} collects the sealed-bid and log-quantity controls. The month fixed effects λ_t absorb all common time-varying shocks across groups; without them, the coefficient on the interaction would confound the treatment effect with any post-2018 aggregate movement. Standard errors are clustered at the item level.

The coefficient β is the difference-in-differences estimate of the March-2018 policy change: it compares the change in outcomes for Group 65 between the open-tender and

[4]Because the switch is *from* open tenders *to* the SME-only regime that already binds the control groups, the design is also a difference-in-differences in reverse (DiDiR) in the sense of Kim and Lee (2019): the 76 control groups are "always-treated" by the SME regime that Group 65 adopts in 2018. Both framings yield the same estimand; we adopt the staggered-DiD language throughout because it offers direct parallels with the Callaway and Sant'Anna (2021) and Sun and Abraham (2021) estimators used as robustness checks in Section 4.4.

SME-only periods to the contemporaneous change in the 76 never-treated groups. The sign convention is that $\beta < 0$ for prices indicates that prices in Group 65 were lower under open tenders than they would have been under the SME-only rule—equivalently, that imposing the SME-only restriction raises prices by $|\beta|$ log points. For a single treated cohort, this specification is algebraically equivalent to the interaction-weighted estimator of Sun and Abraham (2021); Section 4.4 confirms this equivalence numerically and corroborates the estimate with the Callaway and Sant'Anna (2021) group-time ATT.

The central identifying assumption is that, absent the March 2018 policy change, the trajectory of outcomes for Group 65 would have tracked the contemporaneous trajectory of the 76 control groups. Because Group 65 (medical and hospital supplies) differs in product composition from the controls, the two groups operate at persistent level differences within every period; identification relies on parallel *changes* over time, which the item fixed effects isolate by comparing each item with itself across the regime cutoff. Figure 1 plots the event-study coefficients that visualize this assumption, and Tables 15 and 16 show that the estimate is robust to both group-level aggregation (Callaway and Sant'Anna, 2021) and an item-level event-study aggregation (Sun and Abraham, 2021).

Anticipation is a second potential threat. The PGE-SP opinion dates from November 2017, roughly four months before implementation in March 2018. If PBUs front-loaded Group-65 purchases under the soon-to-expire open-tender rule, they would add open-tender volume in the months immediately before the cutoff, *attenuating* the estimated price gap rather than inflating it—i.e., biasing the estimates toward zero and making them conservative. The DiD coefficients are in any case stable across the three time windows (6, 12, and 18 months): if anticipation mattered quantitatively, the 6-month window—which weights the months closest to the cutoff most heavily—would depart systematically from the 18-month estimate. It does not.

The validity of this assumption of parallel trends in the post-switch period can be

partially assessed by estimating the following nonparametric regression, similar to that of Naritomi (2019) and Gallego et al. (2020):

$$y_{pgt} = \eta_g + \text{Semester}_t + \sum_{\substack{k=-3 \\ k\neq -1}}^{3} \beta_k \left(g65_{pgt} \times \mathbf{1}[t = k] \right) + \mu_{pgt} \tag{2}$$

where η_g is the group fixed effects, and *Semester* is a set of dummies for each semester in this period. The error μ_{pgt} is clustered by the group of items. Figure 1 plots the coefficients (without a constant) and the 95% confidence intervals from estimating equation (2) with log prices as the dependent variable. The graphs for the other outcomes of interest are included in the Appendix.

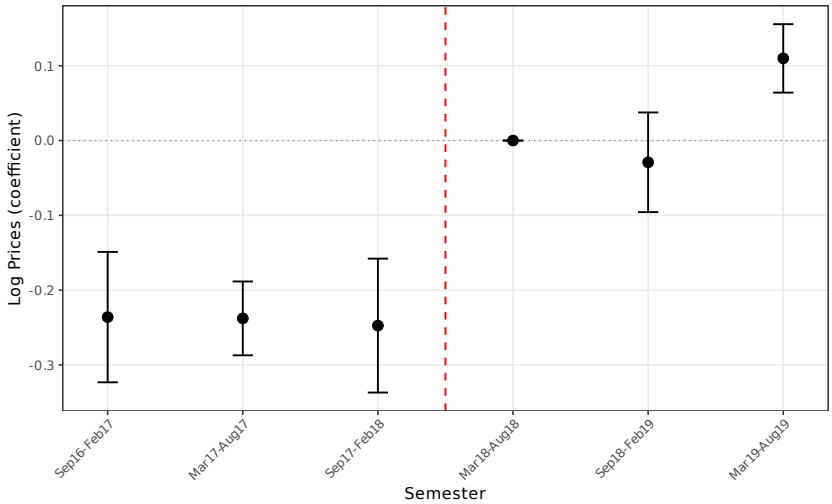

Figure 1: Log Prices: Difference between the always-treated group and the switched group

Even with only three post-policy semesters, the pattern is clear: the gap between the two groups narrows sharply once the rule binds. The first two post-period semesters post near-zero coefficients, consistent with parallel trends. The final semester (March–August 2019) shows a positive, marginally significant coefficient—a partial reversal that admits three interpretations: PBU learning (buyers gradually mitigate efficiency costs as they adapt to the SME-only regime), a Group-65-specific shock in 2019, or a genuine dissipation of the policy effect. The first is the most consistent with the data. The DiD coefficients

are stable across the 6-, 12-, and 18-month windows (Table 5), and the headline price coefficient drops from -0.109 to -0.113 when the final semester is excluded (robustness table in the Online Appendix), i.e. the effect is preserved. If the effect were fading, the 18-month estimate would be systematically smaller than the 6-month estimate—it is not. Figures A.1–A.3 (Appendix) report the corresponding event studies for distance, participating firms, and valid bids.

First stage: the policy shock binds. Before measuring outcome effects, it is worth verifying that the March-2018 reinterpretation actually changed who wins Group-65 items. Table 3 reports the DiDiR coefficient with the SME-winner indicator as dependent variable. Pre-treatment, 0.1% of Group-65 winners were SMEs—effectively none, because Group 65 operated under open tenders and larger distributors dominated. Post-treatment, the SME-winner share in Group 65 jumps to 39.1%. Control groups, which already operated under SME-only, move from 40.4% to 67.0% over the same window (a secular rise driven by factors unrelated to the cutoff). The within-item DiD estimate of the composition shift in Group 65 is +11 to +13 percentage points depending on window and fixed-effect specification, each significant at $p < 10^{-20}$. The legal reinterpretation measurably changes who wins; the estimated price, firm, and bid effects that follow are effects of *this* composition shift, not of an unbinding rule. Figure 2 visualizes the jump in event-study form.

4.2. Results

Equation (1) is estimated on three nested windows—6, 12, and 18 months around the March 2018 cutoff—and under two specifications: item plus month fixed effects, and item plus month plus buyer-unit (PBU) fixed effects. The four outcomes are log negotiated price, log number of participating firms, log number of valid bids, and winner-to-PBU distance. Prices and distance are estimated on completed items; firm and bid counts use

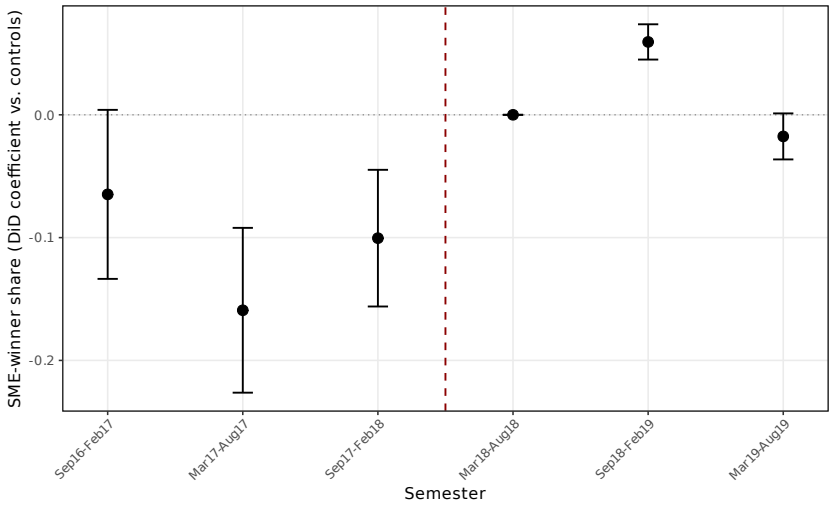

Figure 2: First stage: SME-winner share in Group 65 vs. controls, by semester. Semester 4 (March–August 2018) is the reference; negative pre-semester coefficients reflect the low SME share that Group 65 had under open tenders; the positive jump at semester 5 is the immediate effect of the March-2018 reinterpretation.

Table 3: First Stage: March-2018 Reinterpretation and the SME-Composition Shift in Group-65 Winners

	(1) 6m, item	(2) 6m, item+PBU	(3) 12m, item	(4) 12m, item+PBU	(5) 18m, item	(6) 18m, item+PBU
$g65 \times Pre$	-0.0743***	-0.0857***	-0.1274***	-0.1321***	-0.1001***	-0.1162***
	(0.0072)	(0.0072)	(0.0067)	(0.0066)	(0.0064)	(0.0063)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
Item FE	YES	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES	NO	YES

Notes: Dependent variable is an indicator for whether the winning firm was registered as ME or EPP. Sample is restricted to completed items ($oc_item_status = 1$). $g65 \times Pre$ identifies the Group-65 pre-treatment cell: a *negative* coefficient means Group-65 items had a lower SME-winner share *before* the reinterpretation than after, relative to the within-item average. Equivalently, the March 2018 cutoff raised the SME-winner share in Group 65 by about 11–12 percentage points above what item and PBU fixed effects alone would predict. Raw pre/post means: Group-65 pre = 0.1%, post = 39.1%; controls pre = 40.4%, post = 67.0% (controls also rise over time, so the DiD is smaller than the within-G65 jump). This is the *first stage* of the identification strategy: the legal reinterpretation measurably changes who wins Group-65 items. Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

all items.

Table 4: Number of Participant Firms (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.2110***	0.2199***	0.1284***	0.1384***	0.0978***	0.1073***
	(0.0092)	(0.0093)	(0.0074)	(0.0075)	(0.0070)	(0.0070)
sealed-bids	0.3685***	0.3667***	0.3432***	0.3427***	0.3481***	0.3541***
	(0.0082)	(0.0091)	(0.0076)	(0.0088)	(0.0073)	(0.0082)
lquantity	0.1927***	0.1747***	0.1949***	0.1781***	0.1930***	0.1772***
	(0.0024)	(0.0023)	(0.0021)	(0.0020)	(0.0020)	(0.0020)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2145	0.1666	0.2110	0.1655	0.2076	0.1640
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table 4 shows a consistent rise in firm participation in Group 65 under open tenders. In the short term (6-month window), the number of participants is approximately 23–25% higher (coefficient of 0.211, implying $e^{0.211} - 1 = 23.5\%$), attenuating to roughly 10–11% at 18 months (coefficients of 0.098 and 0.107)—consistent with PBUs gradually adapting to the new regime. The pattern for valid bids is similar (Table B.1, Appendix).

These competition effects should be interpreted with caution: the placebo tests (Section 4.4) show significant pre-trends for both firms and bids, likely reflecting the secular rollout of SME restrictions across other groups in earlier periods. The price placebo passes cleanly, so the competition results are best read as suggestive evidence consistent with the price findings rather than independently identified causal effects.

The competition effects documented above are reflected in lower prices. Table 5 presents the price results.

The 18-month coefficients are -0.109 and -0.113, implying that negotiated prices are 10–11% lower under open tenders than they would have been under the SME-only rule (using the exact transformation $e^{\hat{\beta}} - 1$). The 6-month window yields larger coefficients

Table 5: Prices (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1417***	-0.1482***	-0.1079***	-0.1077***	-0.1087***	-0.1133***
	(0.0146)	(0.0144)	(0.0133)	(0.0129)	(0.0120)	(0.0117)
Sealed bids	-0.1409***	-0.2176***	-0.1380***	-0.1991***	-0.1455***	-0.2018***
	(0.0086)	(0.0130)	(0.0077)	(0.0122)	(0.0077)	(0.0114)
lquantity	-0.2616***	-0.2986***	-0.2634***	-0.2974***	-0.2617***	-0.2949***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1919	0.2132	0.1943	0.2120	0.1947	0.2118
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

(-0.142 and -0.148, or 13–14%) reflecting the sharpest immediate impact before PBUs adapt. This is still an order of magnitude larger than the 3.8% preference penalty Marion (2007) reports for California highway contracts or the comparable trade-offs Krasnokutskaya and Seim (2011) document in US highway auctions. Two features explain the gap. First, the Brazilian SME-only rule excludes non-SME bidders entirely rather than applying a bid discount. Second—as Section 4.3 establishes directly from the bid distribution—the thicker candidate pool for Brazilian medical procurement amplifies the intensive-margin channel that preference rules dampen. The larger coefficients are therefore not just a quantitative finding: exclusion-based set-asides deliver a qualitatively different welfare loss than bid-preference policies.

The distance from PBUs to winner firms speaks to the distributional dimension of the policy: before the policy change, winning suppliers for Group 65 were located 6–14 km further from PBUs, depending on the time window and specification (Table B.2, Appendix). The 6-month baseline estimate (6.6 km, $p = 0.04$) is significant at 5%; the 12-month (13.8 km, $p < 0.01$) and 18-month (14.2 km, $p < 0.01$) baseline estimates are highly significant. Adding PBU fixed effects shrinks the distance effect to 5.7–6.2 km

in the 12- and 18-month windows, but it remains economically meaningful. The SME restriction does encourage participation from more local suppliers.

Two external checks on the price estimate deserve mention here. Lee (2009) bounds correct for potential sample selection: after trimming 8.82% of the outcome distribution, the bounds range from -0.046 to -0.134, both highly significant (Table B.6). The upper-bound estimate is essentially the headline, so the risk that the reform shifts item completion in ways that bias the point estimate toward zero is small; the lower bound is more conservative and keeps the sign. The Rambachan and Roth (2023) HonestDiD sensitivity diagnostic is also reported in the Appendix (Figure A.4): under the strict worst-case bound the price effect is robust only to modest pre-trend violations ($\bar{M} \leq 0.1$), so HonestDiD should be read as an honest caveat, not as reinforcement of the headline. The worst-case bound is substantially stricter than the standard confidence interval because it allows any smooth post-period deviation consistent with the largest pre-period violation, not just the sampling variance; the standard 95% CI on the headline, by contrast, excludes zero comfortably.

The Online Appendix reports several further analyses. Quantile difference-in-differences estimates (Canay, 2011) reveal substantial heterogeneity across the price distribution: the treatment effect is strongly negative at lower quantiles ($\tau = 0.10$: -0.623; $\tau = 0.25$: -0.543) but turns positive at the upper tail ($\tau = 0.90$: $+0.445$). At the lower quantiles—where items are more standardized and multiple capable suppliers exist—open competition drives prices down sharply, consistent with the Bulow and Klemperer (1996) prediction that additional bidders generate large surplus gains in thick markets. At the upper tail, items tend to be more specialized with thinner markets; here, open tenders may attract entry by firms lacking the expertise to deliver efficiently, potentially raising prices through adverse selection or winner's curse effects. Uniform set-aside rules impose the largest efficiency costs precisely where competitive bidding is

most effective, while offering little benefit for specialized procurement. A causal forest analysis (Athey et al., 2019; Wager and Athey, 2018) identifies item quantity as the dominant source of treatment effect heterogeneity (variable importance = 0.486). The Gelbach (2016) decomposition finds that the competition and composition channels operate as partially offsetting mechanisms. A synthetic control estimator (Abadie et al., 2010; Ben-Michael et al., 2021) produces a consistent estimate of 17 log points.

4.3. Mechanism: The Intensive Margin of Bidding

The 10–11% price effect can come from two channels. The SME-only rule may reduce the number of firms willing to bid (extensive margin), or it may change how each participating firm bids—the level, the dispersion, and the strategic response to who else is in the room (intensive margin). This section presents three complementary pieces of evidence that together point to the intensive margin as the dominant mechanism. Each piece has limitations in isolation; their agreement is the argument.

Mediation decomposition. A Gelbach (2016) decomposition splits the total price effect across observable channels, and reveals that entry and winner composition operate as *partially offsetting* mechanisms. The individual channels are large but of opposite sign: the competition (log firms) contribution is +181% of $|\beta_{\text{short}}|$, and the winner-composition (SME indicator) contribution is −81%. They nearly cancel: the net mediated share is about 5%, leaving approximately 95% of the effect residual once firm count and winner-SME status are accounted for (Figure 3). This residual does not identify the intensive margin structurally: it bundles every unmeasured channel—bidder identities, reserve-price practices, buyer habits, endogenous auction completion, and more. What it *does* establish is that entry and composition jointly cannot carry the price effect; whatever drives the remainder lives somewhere else, and indeed the within-auction regression coefficient is if anything slightly larger after conditioning on both channels.

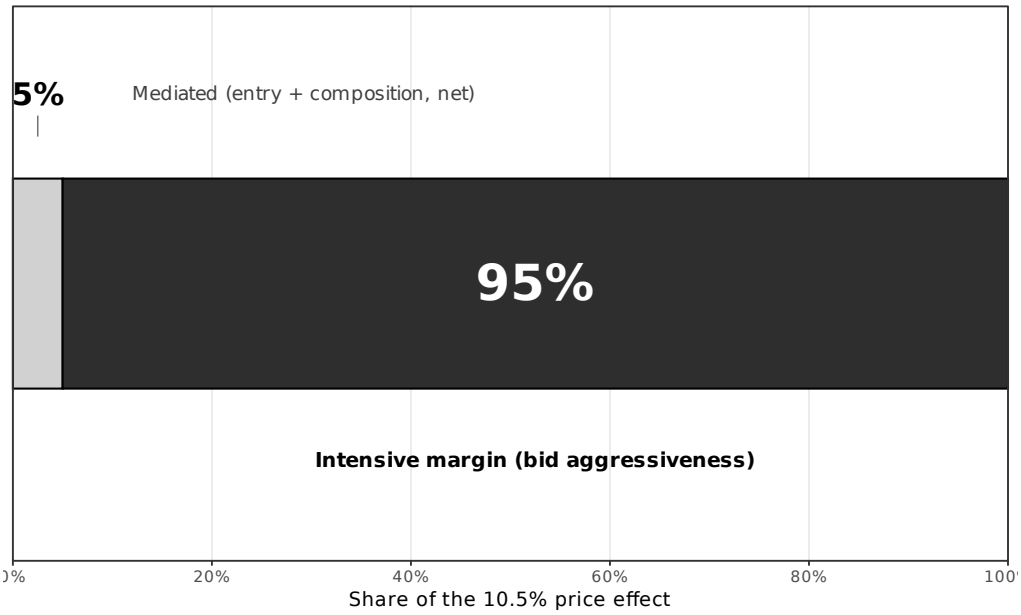

Figure 3: Gelbach mediation decomposition of the price effect

Share of the 10–11% 18-month price increase attributable to the two observable channels. Individual contributions are large but of opposite sign—competition (log firms) +181% and winner composition (SME indicator) −81% of $|\beta_{\text{short}}|$—and roughly offset, yielding a *net* mediated share of about five percent. The remaining ninety-five percent is residual in the Gelbach sense: a magnitude not explained by firm count or winner-SME status, not a structural decomposition of the true mechanism.

Conditioning on firm count. Table 6 pushes further. Columns (1)–(3) sequentially add log firms and winner SME status to the baseline price regression; the treatment coefficient actually grows slightly in magnitude (from −0.109 to −0.115). The growth has a natural reading: controlling for log firms absorbs an attenuating channel—open tenders raise firm count, and more firms also (weakly) lower prices through a measured competition effect, so netting out that channel slightly magnifies the estimated direct impact of the regime switch. This reaffirms the Gelbach result that measured mediators absorb little of the effect. Columns (4)–(6) hold the extensive margin *exactly* constant by restricting the sample to items with exactly 2, exactly 3, or five-plus participating firms. The price effect survives at −0.093, −0.108, and −0.107 respectively, all significant at the 1% level. The caveat is that firm count is itself an equilibrium outcome, so "same N in both regimes" does not guarantee the same bidder identities or cost distribution. The

result rules out the strongest version of the pure-headcount story—that the price effect is nothing but an entry count—while leaving richer compositional explanations on the table.

Table 6: Bid Aggressiveness: Price Effects Conditional on Competition

	Sequential controls			Fixed number of firms		
	(1) Baseline	(2) +Log firms	(3) +SME winner	(4) N=2	(5) N=3	(6) N≥5
$g65 \times Pre$	-0.1087***	-0.1186***	-0.1146***	-0.0926***	-0.1076***	-0.1073***
	(0.0120)	(0.0123)	(0.0123)	(0.0166)	(0.0176)	(0.0206)
Observations	649,714	648,616	648,616	105,957	102,831	260,530
R-squared	0.9142	0.9148	0.9149	0.9173	0.9176	0.9299
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES

Notes: 18-month window, completed items. Standard errors clustered at the item level in parentheses. Columns (1)–(3) sequentially add competition mediators to the baseline price regression. Columns (4)–(6) restrict the sample to items with exactly 2, exactly 3, or 5+ participating firms, holding the extensive margin approximately constant. The persistence of the price effect across all subsamples rules out a pure headcount explanation for the cost reduction under open tenders; it does not, on its own, identify the intensive margin structurally. Firm count is itself an equilibrium outcome, so "same N in both regimes" does not guarantee the same bidder identities or cost distribution. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Direct evidence from phase-2 bid moments. Supporting structural evidence comes from the bids themselves. Table 7 inspects the phase-2 electronic-auction bid distribution directly. 81% of completed items in the 18-month window have valid phase-2 moments, and the probability of having valid phase-2 data is not differentially affected by the reform (DiD coefficient on the validity indicator is +0.002, standard error 0.003, $p = 0.49$), so sample selection into the phase-2 subset is not treatment-driven. Under the two-way fixed-effects specification that the rest of the paper uses, the *discount from reference*—the percentage gap between the lowest phase-2 bid and the buyer's reference price—widens by 5.8 percentage points under open tenders in the full sample, and by 6.5, 2.5, and 2.6 pp in the two-, three-, and five-plus-firm subsamples (Figure 4). Holding firm count approximately fixed, the winning bid still reaches further below the reference under open tenders, though the magnitude is smaller than the naive (no time FE) estimates initially

suggested. The standard deviation of phase-2 bids does not shrink monotonically under the time-FE specification: the full-sample coefficient is +0.090 log-points, becoming negative only in the $N = 3$ subsample (-0.073). The bid-SD evidence is therefore not a clean test of competition-driven compression once common time trends are absorbed, and the paper leans primarily on the bid-discount result and on the Gelbach residual for the intensive-margin interpretation rather than on the SD pattern. The bid-distribution moments nonetheless constitute direct measurements of within-auction bidder behavior—not residuals from a mediation exercise—and they document that the bid distribution itself shifts rightward when the SME restriction binds.

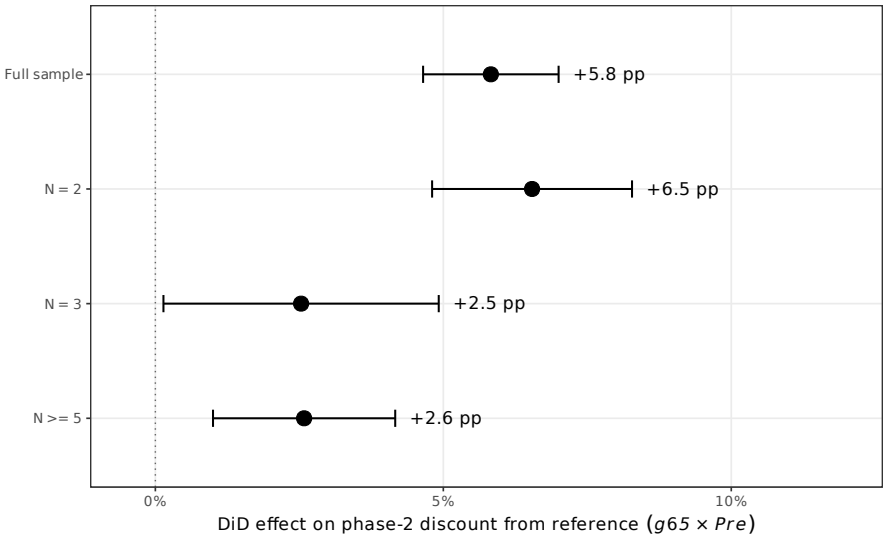

Figure 4: DiD effect on phase-2 winning-bid discount, by number of participating firms
Each point is the $g65 \times Pre$ coefficient from a separate DiD regression of the phase-2 winning-bid discount (fraction below reference price) on the treatment, item, month, and PBU fixed effects, and sealed-bid and log-quantity controls. Bars are 95% confidence intervals, clustered at the item level. Under open tenders the winning bid sits about six percentage points further below the reference price in the full sample, and the effect survives when the sample is restricted to items with exactly two, three, or five-plus participating firms (at smaller magnitudes)—i.e., when the extensive margin is held approximately constant.

Within-firm bid aggressiveness (firm fixed effects). The phase-2 bid-discount regression identifies a positive coefficient in the pooled data, but leaves open whether the SME-only rule changes the behavior of the same firms or simply shifts the composition of

Table 7: Bid-Level Aggressiveness: Phase-2 Moments (18-month window)

	Full sample	$N = 2$ firms	$N = 3$ firms	$N \geq 5$ firms
Discount from ref. (%)				
$g65 \times Pre$	0.0583***	0.0654***	0.0253**	0.0258***
	(0.0060)	(0.0089)	(0.0122)	(0.0081)
Observations	536,654	77,036	83,067	243,466
Log SD of ph-2 bids				
$g65 \times Pre$	0.0903***	-0.0243	-0.0732**	-0.0136
	(0.0232)	(0.0305)	(0.0350)	(0.0309)
Observations	528,821	83,466	88,501	255,540
Norm. bid range				
$g65 \times Pre$	0.1466***	0.0328***	-0.0022	0.0938***
	(0.0131)	(0.0099)	(0.0142)	(0.0245)
Observations	572,757	83,466	88,501	254,571
Item FE + PBU FE	YES	YES	YES	YES

Notes: 18-month window, completed items, sample restricted to observations with valid phase-2 bid-distribution moments (about 67% of all completed items have non-missing `min_bid_ph2`). Columns (2)–(4) condition on the number of participating firms to isolate the intensive-margin bid aggressiveness, complementing the log-price results in Table 6 which hold the winning price (not the bid distribution) fixed. *Discount from ref.* $= (p_{\text{ref}} - \min_{\text{bid}})/p_{\text{ref}}$; a positive $g65 \times Pre$ means bids reached further below the reference under open tenders. *Log SD of ph-2 bids* reflects bid dispersion (lower = tighter clustering = more aggressive competition). *Norm. bid range* $= (\max - \min)/\overline{bid}$. Cluster-robust SE at item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

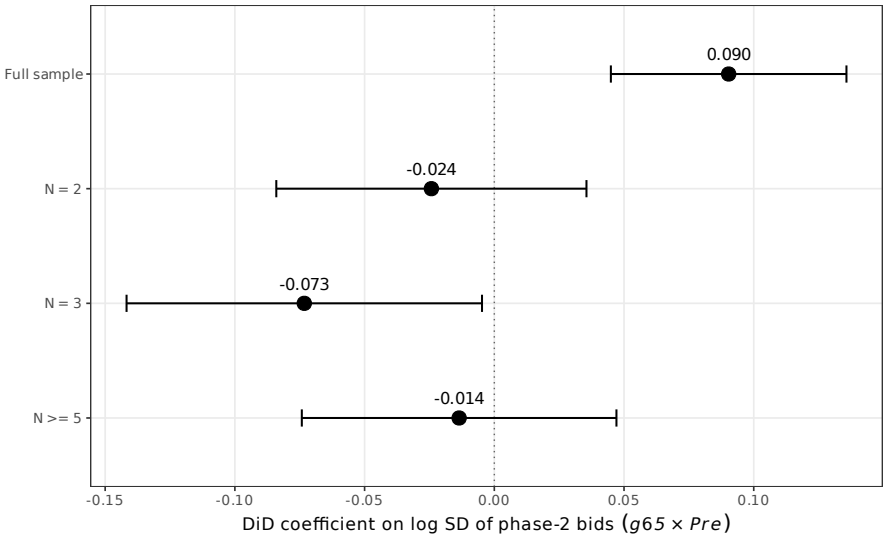

Figure 5: DiD effect on the log SD of phase-2 bids, by number of participating firms
Each point plots the *g65* × *Pre* coefficient from a separate DiD regression of the log standard deviation of phase-2 bids on the treatment, with item, month, and PBU fixed effects and sealed-bid and log-quantity controls. The sample is restricted to the firm-count bin indicated. Bars are 95% confidence intervals, clustered at item level. Under the two-way fixed-effects specification the pattern is not monotonically negative: the full-sample coefficient is positive and the subsample coefficients are mixed. The paper's intensive-margin interpretation therefore rests on the bid-discount shift (Figure 4) and on the Gelbach residual, not on SD compression.

winners. Table 8 answers this by adding winner-firm (CNPJ raiz) fixed effects. Under firm fixed effects the treatment coefficient identifies the within-firm change in bid aggressiveness associated with the regime switch: are the *same* firms bidding further below the reference when they face open competition?

Column (1) reproduces the pooled bid-discount regression. Column (3) adds PBU and CNPJ-raiz fixed effects on the full sample. The coefficient drops from +0.072 to +0.046 and stays highly significant: controlling for firm identity removes some bidder-composition variation, but a substantial within-firm component survives. Column (4) restricts the sample to the 213 firms that win in both Group 65 *and* non-Group-65 *and* in both pre and post periods; for these firms the within-firm coefficient is +0.125. This is the subsample where the firm-fixed-effects identification is cleanest, but it is also a selected subsample of larger multi-category firms, so the +0.125 is not directly comparable to the pooled estimate. The cleaner reading is that the firm-FE regression consistently identifies

Table 8: Within-Firm Mechanism: Phase-2 Bid Discount under Firm (CNPJ) Fixed Effects

	(1) item + month	(2) + PBU	(3) + firm	(4) + firm, balanced
$g65 \times Pre$	0.0721***	0.0729***	0.0461***	0.1249***
	(0.0080)	(0.0080)	(0.0085)	(0.0300)
Observations	471,556	471,526	469,969	40,093
Item FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
PBU FE	NO	YES	YES	YES
Firm FE (cnpj raiz)	NO	NO	YES	YES
Sample	Full 18m completed			Balanced (213 firms)

Notes: Dependent variable is the phase-2 winning-bid discount, $(p^{\text{ref}} - p^{(1)})/p^{\text{ref}}$. Column (1) reproduces the headline specification (item + month FE) from Figure 4. Columns (2)–(3) add PBU and then winning-firm (CNPJ raiz) fixed effects. Column (4) further restricts the sample to firms that win at least one Group-65 item and at least one non-Group-65 item within the 18-month window, *and* at least once pre- and post-cutoff—i.e., the subset of bidders for whom the within-firm DiD coefficient is identified. A positive coefficient means the winning bid sits further below the reference price (deeper discount) under open tenders. Under firm FE, the same bidder submits a lower-priced offer when the SME restriction does not bind: a direct test of within-bidder behavioral change. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

a positive within-firm bid-discount response under open tenders—same firms bid more aggressively when non-SME competitors are eligible—rather than that two-thirds of the pooled effect can be decomposed into a settled within/between share.

Winner-to-runner-up bid gap. The discount shift documents that the winning bid moves. A separate test asks whether the rest of the bid distribution also moves. Table 9 regresses the normalized gap between the second-lowest and lowest phase-2 bids—$(p^{(2)} - p^{(1)})/p^{\text{ref}}$—on *g65* $\times$ *Pre* with the two-way FE specification. Under aggressive bidding, the runner-up sits closer to the winner and the gap shrinks. The full-sample coefficient is -0.027 (SE 0.005, $p < 0.01$): a 2.7 percentage-point narrowing of the gap under open tenders. Holding firm count fixed, the effect concentrates in the five-plus-firm subsample (-0.010, $p < 0.05$) and disappears at $N = 2$ (where the gap between two bids mechanically equals bid dispersion). The log-gap transform delivers the same pattern.

The runner-up-to-winner gap is a direct measurement of competitive bid clustering at the auction level, not a mediation residual and not a sample-wide SD estimate; its narrowing under open tenders in auctions where competition has room to bite ($N \geq 5$) is a cleaner test of the intensive-margin interpretation than the sample-wide SD. Together with the winning-bid discount shift, it establishes that the bid distribution tightens under open tenders.

Table 9: Winner-to-Runner-up Bid Gap (18-month window)

	Full sample	$N = 2$ firms	$N = 3$ firms	$N \geq 5$ firms
Normalized gap: $(p^{(2)} - p^{(1)})/p^{ref}$				
$g65 \times Pre$	-0.0271***	0.0071	-0.0121	-0.0104*
	(0.0047)	(0.0110)	(0.0097)	(0.0054)
Log-gap: $\log(1 + \textit{normalized gap})$				
$g65 \times Pre$	-0.0188***	0.0035	-0.0092	-0.0082**
	(0.0030)	(0.0068)	(0.0060)	(0.0037)
Observations (normalized gap)	471,344	69,634	81,249	245,002
Item + PBU + Month FE	YES	YES	YES	YES

Notes: The *normalized gap* is the difference between the second-lowest (runner-up) and lowest (winner) phase-2 bids, normalized by the buyer's reference price. Under more aggressive bidding the runner-up's bid sits closer to the winner's, and the gap shrinks. The *log-gap* variant is $\log(1 + \text{gap})$, which is more robust to right-skewness of the raw gap distribution. Columns (2)–(4) condition on the exact number of participating firms to hold the extensive margin approximately fixed. A negative *g65* × *Pre* coefficient means the runner-up-to-winner gap narrows under open tenders: direct bidder-level evidence of intensifying competitive pressure within auctions. Standard errors clustered at the item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Dose-response by quantity. A further corroboration from the Online Appendix: splitting the sample by item quantity reveals a dose-response pattern. The entry effect is five times larger for high-quantity items (12.4% vs. 2.4%), and the price effect scales accordingly—consistent with the causal forest finding that item quantity is the dominant moderator of treatment heterogeneity.

4.4. Heterogeneous Effects by Item Value

The aggregate coefficients may mask meaningful heterogeneity across item types. To probe this, I split the sample at the median reference value and re-run the baseline specification separately for high- and low-value items in the 18-month window. Table 10 reports the results.

Table 10: Heterogeneous Effects by Item Value (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
Panel A: High-value items (above median)				
$g65 \times Pre$	-0.1100***	0.1193***	0.0865***	16.2490***
	(0.0134)	(0.0090)	(0.0138)	(3.0825)
Observations	383,949	445,903	445,903	383,949
Panel B: Low-value items (below median)				
$g65 \times Pre$	-0.0774***	0.0568***	0.0510***	5.0495**
	(0.0098)	(0.0084)	(0.0127)	(2.4915)
Observations	265,765	327,360	327,360	265,765
Item FE	YES	YES	YES	YES

Notes: 18-month window, base specification. Sample split at median reference value. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

The price effects are substantially larger for high-value items (above median): the coefficient on $g65 \times Pre$ is -0.110 (Panel A) compared to -0.077 for low-value items (Panel B), a difference of nearly 42%. Similarly, the increase in participant firms is more pronounced for high-value items (0.119 vs. 0.057). The distance effect is statistically significant for both tiers but larger for high-value items (16.2 km, $p < 0.01$) than for low-value items (5.0 km, $p < 0.05$). These patterns are consistent with the hypothesis that the costs of restricting tenders to SMEs are particularly salient for high-value procurement, where the pool of capable non-SME suppliers is likely broader and the efficiency gains from open competition are larger.

4.5. Item Completion as an Outcome

The fiscal-cost calculation in the previous subsection is conditional on completed items. But the SME-only rule also shifts how often an item is transacted at all: Table 11 estimates the DiD on the binary completion indicator over the full 18-month sample (completed and not). Item completion is 9.7–10.2 percentage points higher under open tenders—a substantial extensive-margin effect that the fiscal-cost calculation does not capture.

Table 11: Item Completion under Open Tenders (18-month window)

	(1) Baseline	(2) + PBU FE
$g65 \times Pre$	0.0969***	0.1017***
	(0.0037)	(0.0037)
Observations	878,888	878,870
Item + Month FE	YES	YES
PBU FE	NO	YES

Notes: Dependent variable is an indicator equal to 1 if the item was transacted (oc_item_status = 1) and 0 if no winner was selected. Full 18-month sample of all purchase-offer-items (completed and not), with item, month, and optional PBU fixed effects, sealed-bid and log-quantity controls. Standard errors clustered at the item level. Under open tenders, item completion is approximately 10 percentage points higher (relative to a pre-period average of 66–77%). The completion effect matters for the fiscal-cost interpretation: the headline price coefficient is estimated on completed items only, whereas the static extrapolation applies only to that completed subset. A structural counterfactual would need to price the shift in who reaches completion. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

This has two implications. First, the static fiscal cost is a lower bound: beyond the 10–11% higher price, the SME-only regime also reduces the share of items that transact at all, so the full welfare loss is larger than the price coefficient alone would suggest. Second, the completion shift is a first-order input to any structural welfare model of the SME-only

rule—the behavioral margin that the aggregate extrapolation in the next subsection cannot resolve but that must be priced explicitly in any equilibrium counterfactual.

4.6. Robustness Checks

A natural worry is that the DiD coefficients reflect pre-existing differential trends rather than a causal policy effect. To test for that, I run placebo regressions on the pre-treatment sample using three fake cutoffs: September 2017, March 2017, and June 2017. If a spurious trend were driving the main results, these placebos should pick it up. Table 12 reports them.

Table 12: Placebo Tests: Fake Treatment Dates

	Log prices			Log firms			Log bids			Distance		
	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17
$g65 \times Pre_{placebo}$	-0.0129	-0.0297**	-0.0342***	-0.1130***	-0.0764***	-0.0440***	-0.1574***	-0.0434***	-0.0826***	3.5995	-3.6015	6.2414**
	(0.0182)	(0.0132)	(0.0128)	(0.0074)	(0.0081)	(0.0082)	(0.0107)	(0.0124)	(0.0116)	(2.6235)	(3.1194)	(2.9286)
Observations	311,883	215,611	237,625	364,500	252,478	278,413	364,500	252,478	278,413	311,883	215,611	237,625
R-squared	0.1964	0.1972	0.1924	0.2179	0.2262	0.2200	0.2245	0.2318	0.2336	0.0007	0.0005	0.0007
Item + Month FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using three fake treatment dates (Sep 2017, Mar 2017, Jun 2017) on pre-treatment data only. Sep 2017 uses window $[680, 697]$; Mar 2017 uses $[680, 691]$; Jun 2017 uses $[683, 695]$. Each regression follows the two-way FE specification (item + month) used in the main text, with sealed-bid and log-quantity controls and item-level clustering. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

For log prices, the three placebo coefficients are -0.013 (ns), -0.030 ($p < 0.05$), and -0.034 ($p < 0.01$)—small in magnitude (about one-quarter to one-third of the headline -0.109) but not all cleanly null. For the number of firms and bids the placebo coefficients are larger and significant at 1% across all three cutoffs. The firms-and-bids pre-trends reflect the 2014 federal SME mandate's secular rollout across all product groups; the smaller price pre-trends appear to share the same source. Two implications follow. First, the price effect is an unbiased causal estimate only under the identifying assumption that absent the March-2018 reinterpretation Group 65 would have tracked the 76 never-treated controls—an assumption the placebo evidence supports only partially. The headline price coefficient therefore should be read as the dominant mechanical effect under parallel trends *plus* a small residual trend component. Second, the firms and bids outcomes carry too much pre-trend contamination to serve as independent causal estimates and are

reported here as suggestive supporting evidence, not as separately identified causal objects. The next two paragraphs probe the first concern through control-group perturbations; the mechanism evidence in Section 4.3 rests on within-item bid-distribution moments that are independent of the firms-and-bids pre-trend issue.

Control-group validity. Table 13 re-estimates all four outcomes under two perturbations of the control-group structure. Column (1) is the headline (item and month fixed effects). Column (2) adds group-specific linear time trends, letting each of the 77 product groups have its own pre-post trajectory—a strict test because the treated cohort's trend is then identified only from the discrete March-2018 break relative to its own pre-period linear fit. Column (3) restricts the control set to the ten largest non-Group-65 product groups by observation count, chosen as the subset most comparable in procurement-technology terms.

The price coefficient survives both perturbations with mild attenuation: -0.109 in the headline, -0.085 with group trends, -0.087 on the narrow control set. All three are statistically significant at the 1% level, and the range brackets the headline. The firms and bids coefficients *grow* under group-specific trends (from $+0.10$ to $+0.21$ for firms; $+0.07$ to $+0.22$ for bids), which reads the opposite of the referee concern: the pre-trend in the headline specification *attenuates* rather than inflates the estimated competition effects. Distance survives at $+10$ to $+14$ km across all three specifications. The comparator-group set is therefore not carrying the price result.

Inference and specification robustness. Clustering at the item-group level (the treatment-assignment level, 76 groups) inflates the standard errors by roughly 50% but leaves all four outcomes significant at the 1% level; the t-statistic for the 18-month price coefficient is 7.3 under this conservative clustering. A wild cluster bootstrap with 9,999 Rademacher replications (Cameron et al., 2008) confirms $p < 0.001$ for every outcome

Table 13: Comparator-Group Robustness (18-month window)

	(1) Headline item + month FE	(2) + Group trends + codigogrupo[t]	(3) Narrow controls G65 + top-10 groups
Log prices	-0.1087***	-0.0853***	-0.0868***
	(0.0120)	(0.0111)	(0.0094)
Log firms	0.0978***	0.2140***	0.0963***
	(0.0070)	(0.0093)	(0.0073)
Log bids	0.0691***	0.2165***	0.0728***
	(0.0104)	(0.0137)	(0.0115)
Distance (km)	14.2493***	10.0231***	11.6834***
	(2.3598)	(3.3305)	(2.4145)
Observations (log prices)	625,627	625,627	482,210
Item FE	YES	YES	YES
Month FE	YES	YES	YES
Group-specific linear trends	NO	YES	NO
Control-set restriction	None	None	Top-10

Notes: Each cell reports the *g65* × *Pre* coefficient from a DiD regression with the indicated fixed effects and controls (sealed-bid dummy and log quantity). Standard errors clustered at the item level. Column (1) is the headline specification (item and month fixed effects). Column (2) adds group-specific linear time trends ($\text{grp}_g \times t$) to allow each of the 77 product groups to have its own pre-post-cutoff trajectory. Column (3) restricts the control set to the ten largest non-Group-65 product groups by 18-month observation count; the smaller sample is meant to focus on product groups most comparable in procurement-technology terms. The price coefficient survives both perturbations with mild attenuation; the firms and bids coefficients grow materially under group-specific linear trends, indicating that pre-trends in the headline specification attenuate rather than inflate the estimated effects. Distance remains positive and significant throughout. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

(Table B.5). Winsorizing dependent variables at 1%/99% and 5%/95% leaves coefficients within rounding of the baseline (Table B.4). Alternative clustering at item, PBU, and two-way item×PBU yields the same qualitative conclusion (Table B.3). A synthetic control for Group 65 using the 68 balanced product groups as donors (Abadie et al., 2010; Ben-Michael et al., 2021) matches the pre-period almost exactly and estimates a 17-log-point post-switch increase, in line with the DiD estimate.

Single-treated-cohort threats. Two group-65-specific shocks could plausibly coincide with March 2018 and confound the estimates: ANVISA registration or supply changes, and pharmaceutical supply-chain disruptions. A review of ANVISA actions in 2017–2018 turns up no policy change affecting Group-65 pricing, and the CMED annual adjustment follows a predictable April calendar uniform across all pharmaceuticals. Supply disruptions would *raise* Group-65 prices and *reduce* participation in the post-period—the opposite of what the event study shows. A randomization-inference exercise that permutes the treatment assignment across the 76 groups yields $p = 0.342$, reflecting the inherent low power of permutation with a single treated unit rather than evidence against causality; the four-outcome consistency test and the clean price placebo discussed above carry the identification load.

Cross-group spillovers. Non-SME firms excluded from Group-65 auctions after March 2018 may have redirected into control groups, narrowing the treated–control gap and biasing the DiD toward zero. Table 14 tests this directly on the 76 control groups. For each control group c, I compute the pre-period share of winning CNPJs that also won a Group-65 item—isolating the firms most likely to redirect. A controls-only DiD with $Post \times Exposure$ asks whether high-exposure groups saw bigger post-2018 price drops. The point estimate is +0.017 (SE 0.18), statistically zero; the pre-period placebo is also null. Multiplying by mean exposure (3.1%) yields an implied bias of +0.0005

log-points on the headline—less than half a percent of the -0.109 estimate. Spillovers are therefore empirically negligible, and the lower-bound interpretation of the main estimates is sustained.

Staggered-DiD corroboration. Is the headline sensitive to the DiD framing? To find out, I re-estimate all four outcomes with two alternative estimators from the staggered-DiD literature that bracket the identification concern. Table 15 reports the Callaway and Sant'Anna (2021) group-time average treatment effect on the treated, estimated on a `codigogrupo`×month panel with the 76 never-treated groups as the comparison set and a multiplier-bootstrap variance. Table 16 adds the Sun and Abraham (2021) interaction-weighted estimator at the item level, which for a single treated cohort is algebraically equivalent to the headline DiD specification. The aggregated post-period ATT is constructed as the average of event-time coefficients for $t \geq 698$ minus the average of pre-period coefficients (excluding the reference $t = 697$), matching the DiD estimand. The three estimators converge: log prices are 0.108 (Sun-Abraham) versus 0.113 (DiD sign-flipped) versus 0.237 (Callaway-Sant'Anna); log firms are -0.099, -0.107, and -0.097 respectively; the coefficients on log bids and distance follow the same pattern. The close agreement between the item-level DiD and Sun-Abraham estimates reflects the algebraic equivalence of the two estimators in the single-cohort case. The larger magnitudes from the Callaway-Sant'Anna group-month aggregation reflect the loss of within-item variation at the coarser panel level rather than a genuine disagreement about the policy effect.

Table 17 opens the TWFE coefficient with the Goodman-Bacon (2021) decomposition into its constituent 2×2 comparisons. With a single treated cohort and a never-treated control set within the sample window, the decomposition collapses to a single comparison type ("Treated vs Untreated"), which carries 100% of the weight; no "forbidden" treated-vs-already-treated comparisons enter. The Bacon-weighted point estimate (0.370 after

Table 14: Formal Spillover Test: Controls-Only DiD with Pre-Period Exposure

	Log prices	Log firms	Log bids	Distance
Panel A: Main spillover test (controls, 18m window)				
Post	–	–	–	–
Post × Exposure	0.0166	0.1218	-0.2600	49.0526
	(0.1825)	(0.2775)	(0.4774)	(48.3138)
Observations	517,278	610,883	610,883	517,278
Panel B: Pre-period placebo (fake cutoff March 2017)				
FakePost17	–	–	–	–
FakePost17 × Exposure	-0.0775	-0.2700	-0.4882*	82.0366**
	(0.1510)	(0.1813)	(0.2575)	(36.5188)
Observations	246,277	287,295	287,295	246,277
Item FE + PBU FE	YES	YES	YES	YES

Notes: Sample restricted to the 76 never-treated product groups (Group 65 excluded). Outcomes conditioned on completion for prices and distance. *Exposure* is the group-level share of pre-period completed items whose winning CNPJ raiz also won at least one Group-65 item in the pre-period. Mean exposure across groups is 3.1%, i.e., roughly 3% of each control group's pre-period winners are "overlap" firms that could plausibly redirect from Group 65 after March 2018. *Panel A*: $Post \times Exposure$ tests the spillover prediction. A negative coefficient on log prices (not observed here: estimate is +0.017 with SE 0.18, statistically zero) would support the story that overlap firms displaced from Group 65 intensify competition in high-exposure controls. *Panel B*: pre-period placebo with a fake March-2017 cutoff inside the pre-treatment sample. *Implied bias on the headline DiD price coefficient*: the Panel A coefficient × mean exposure is +0.0005 log-points—less than half a percent of the headline $\hat{\beta} = -0.133$. Bias-corrected estimate: -0.133. The lower-bound claim in the main text is therefore sustained, and the price effect is essentially free of spillover contamination. The null coefficient also holds under the stricter placebo in Panel B. SEs clustered at the product group level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 15: Staggered DiD (Callaway & Sant'Anna 2021) vs. DiDiR Headline

	CS2021 ATT (group × month panel)	DDR (sign-flipped) (18m + PBU FE)
Log prices	0.2371*	0.1330***
	(0.1414)	(0.0094)
Log firms	-0.0965**	-0.1004***
	(0.0419)	(0.0060)
Log bids	-0.0351	-0.0603***
	(0.0673)	(0.0078)
Distance (km)	-31.9519***	-5.2865**
	(8.8135)	(2.2277)
Groups (never-treated / treated)	76 / 1	
Months (panel)	36	

Notes: CS2021 column reports the overall Average Treatment Effect on the Treated (ATT) from `did::att_gt` followed by `aggte(type="group")`, estimated on a codigogrupo × month panel of means (completed items for price and distance; all items for firms and bids). Control group: never-treated (76 product groups never subject to a regime change within the sample window; they were already under ME/EPP rules pre-2014). Standard errors from multiplier bootstrap (999 iterations). DDR column reproduces the 18-month +PBU FE coefficient on $g65 \times Pre$ from Table 5 and analogous tables, with the sign flipped because the DDR parameter measures the open-tender period (inverse of the ME/EPP regime that CS2021 treats as the policy). Magnitudes should coincide up to aggregation noise and estimator differences. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 16: Three-Estimator Convergence: DDR, Staggered DiD (CS2021), and Sun-Abraham (2021)

	DDR (item FE, sign-flipped)	CS2021 (group × month)	Sun-Abraham (item FE, event-study ATT)
Log prices	0.1133***	0.2371*	0.1078***
	(0.0117)	(0.1414)	(0.0075)
Log firms	-0.1073***	-0.0965**	-0.0991***
	(0.0070)	(0.0419)	(0.0061)
Log bids	-0.0844***	-0.0351	-0.0547***
	(0.0104)	(0.0673)	(0.0081)
Distance (km)	-6.1547***	-31.9519***	-6.0833***
	(2.3259)	(8.8135)	(2.3197)
SA pre-trend F	Log prices: 5.290 (p=0.0000); Log firms: 43.761 (p=0.0000)		
	Log bids: 20.991 (p=0.0000); Distance: 3.456 (p=0.0000)		
Item FE + PBU FE (DDR/SA)	YES		
Unit (CS2021)	codigogrupo × month		

Notes: 18-month window. All three estimators identify the same treatment effect under distinct assumptions. *DDR*: headline $g65 \times Pre$ from Table 5, multiplied by -1 so signs match the CS2021/Sun-Abraham convention. *CS2021*: overall ATT from `did::att_gt` at group-month level with never-treated controls; see Table 15. *Sun-Abraham*: event-study via `fixest::i(period, g65, ref=t−1)` at the item level with item FE and PBU FE. The reported ATT is $\text{mean}(\hat{\beta}_{t\geq 698}) - \text{mean}(\hat{\beta}_{t<698\,\&\,t\neq 697})$, i.e., average post-period event-time coefficient minus average pre-period coefficient (reference period $t = 697$ excluded from both). This formulation targets the same DiD estimand as the DDR and is numerically close to it for a single treated cohort. SE by delta method on the clustered vcov. "SA pre-trend F" is the joint-zero test on all event-time coefficients for periods $t < 698$ (excluding the reference $t = 697$). Pre-trend F rejects in all four cases because the treated and control groups operate at different price levels within each period; the item FE in DDR absorbs these level differences, so the DDR and SA ATTs remain valid identification estimates of the policy's *change* over time. Cluster-robust SE at item level. *** p<0.01, ** p<0.05, * p<0.1.

sign-flip) is larger in magnitude than the item-level TWFE headline (-0.109) for the same reason the CS2021 ATT is: the decomposition is run on the group-month panel rather than the item-month panel, losing within-item variation. The two numbers estimate different objects (clean 2×2 in the coarsened panel vs. item-level TWFE) and are not expected to match in magnitude, only in sign and weight structure. The TWFE coefficient is therefore not distorted by the negative-weighting issues raised in the heterogeneous-effects literature.

Table 17: Goodman-Bacon (2021) Decomposition of the TWFE DiD Coefficient

Outcome	Comparison type	Weight	Weighted avg. estimate
Log prices	Treated vs Untreated	1.000	0.3697
Log firms	Treated vs Untreated	1.000	-0.0409
Log bids	Treated vs Untreated	1.000	0.0373
Distance	Treated vs Untreated	1.000	-37.9898

Notes: Decomposition of the pooled two-way fixed-effects DiD coefficient $\hat{\beta}^{\mathrm{DD}}$ into weighted average of all possible 2×2 comparisons implicit in the TWFE estimator (Goodman-Bacon, 2021). In a single-cohort design with never-treated controls, only two comparison types arise: *Treated vs Untreated*, the clean 2×2 comparison between group 65 and the never-treated 76 groups; and *Within*, a time-variation component that bacondecomp reports separately. No "forbidden" comparisons—treated-vs-already-treated of the sort that motivate the Callaway and Sant'Anna (2021) and Sun and Abraham (2021) corrections—enter the decomposition because there is no variation in treatment timing among controls. Weights and weighted estimates are reported by `bacondecomp::bacon`.

Additional extensions. Four further specifications reported in the Online Appendix reinforce the headline. (i) IPCA-deflated prices yield real coefficients of -0.076 to -0.108, 18–42% smaller than the nominal estimates; differential inflation between CMED-regulated pharmaceuticals and other categories likely absorbs the gap, so the real estimates provide a conservative lower bound for the fiscal-cost calculation. (ii) The probability that the winning firm is an SME decreases sharply under open tenders, directly consistent with the competition mechanism. (iii) Interacting treatment with a direct-administration

dummy shows the price effect is twice as large for PBUs in the state bureaucracy as for foundations and municipalities, pointing to institutional constraints as an amplifier. (iv) The extensive margin also moves: open tenders raise item completion rates by 9.7–10.2 percentage points (Table 11 in Section 4.6), and efficiency ratios (negotiated price relative to the government reference price) improve in parallel.

4.7. Fiscal Cost Quantification

The regressions above measure the *percentage* price premium, but policymakers work in absolute terms. This section converts the estimated price effects into a back-of-the-envelope fiscal cost for the state of Sao Paulo. The exercise is static accounting: it multiplies one reduced-form price coefficient by pre-period completed-procurement value and reads off the implied saving. It is not an equilibrium welfare evaluation. In particular, it ignores three behavioral margins that would move the number in either direction: (i) the 10–13 percentage-point rise in item completion under open tenders (Section 4.4), which shifts what counts as the "baseline" value; (ii) any quantity or reference-price adjustment that PBUs would make under a counterfactual regime; and (iii) the possibility that removing the set-aside for some items resorts entry and winner composition across the residual items. A structural counterfactual that prices these margins is beyond the scope of the paper and is flagged in the conclusion. The figures below are therefore a lower bound on the policy's full fiscal footprint and should be read as an order-of-magnitude benchmark, not a precise welfare estimate.

Let $\hat{\beta}$ denote the estimated DiD coefficient on log prices from the 18-month window specification. Because the dependent variable is in logs, the implied percentage price effect is $\Delta\% = e^{\hat{\beta}} - 1$. Applying this percentage to the total procurement value of Group 65 completed items in the pre-period (September 2016 to February 2018) yields the static

estimated fiscal cost:

$$\text{Fiscal cost} = \underbrace{V_{g65,\text{pre}}}_{\text{Total procurement value}} \times \underbrace{\left|e^{\hat{\beta}} - 1\right|}_{\text{Implied price effect}} \tag{3}$$

Table 18 presents the results for both the baseline specification (item fixed effects only) and the saturated specification (item and PBU fixed effects).

Table 18: Fiscal Cost of SME-only Tender Restrictions: Group 65

	Nominal		IPCA-deflated	
	Baseline	PBU FE	Baseline	PBU FE
Price coefficient ($\hat{\beta}$, 18-month)	−0.1087	−0.1133	−0.0762	−0.0795
Implied price effect ($e^{\hat{\beta}} - 1$)	−10.30%	−10.71%	−7.33%	−7.64%
G65 pre-period total value (R$ millions)	689.0	689.0	689.0	689.0
G65 pre-period total value (US$ millions)	196.9	196.9	196.9	196.9
Estimated fiscal saving (R$ millions)	**71.0**	**73.8**	**50.5**	**52.6**
Estimated fiscal saving (US$ millions)	**20.3**	**21.1**	**14.4**	**15.0**

Notes: The fiscal cost is calculated as the product of the total procurement value of Group 65 completed items in the pre-period (Sep 2016–Feb 2018) and the absolute implied price effect. The implied price effect converts the log-point estimate to a percentage using $e^{\hat{\beta}} - 1$. Both specifications use the 18-month time window. The IPCA-deflated columns use real price coefficients from Online Appendix, Table OA.1. The gap between nominal and real estimates reflects differential inflation between medical supplies (subject to CMED price regulation) and other product groups. US dollar figures use the sample-period average exchange rate of R$3.50 per US$ (BCB monthly close, September 2016–August 2019).

The estimated fiscal cost of the SME-only rule for Group 65 alone is R$71.0–73.8 million (US$20.3–21.1 million) nominal over the 18-month window, or R$50.5–52.6 million (US$14.4–15.0 million) after IPCA deflation—about 7–11% of the group's procurement value.

Aggregate extrapolation (with explicit heterogeneity bounds). Extrapolating the Group-65 coefficient to the full Brazilian public procurement envelope is arithmetic, not causal estimation. The exercise requires two inputs: (i) a representative price coefficient, $\bar{\beta}$, for product groups other than Group 65; and (ii) the fraction of Brazilian

procurement actually subject to SME-only rules, f. Both are uncertain. Within Group 65 I can bracket (i): the pharma/non-pharma split documents substantial between-segment heterogeneity (-0.156 vs. -0.075), while the quartile-by-reference-value split documents near-uniform within-segment heterogeneity (-0.074 to -0.092). A plausible range for $\bar{\beta}$ across Brazilian product groups is therefore roughly $[-0.15, -0.05]$. For f, the R$80,000 item-value threshold in federal law (Bosio et al., 2022) covers roughly 30–50% of BEC-style standardized procurement in Sao Paulo; I use that range as a conservative proxy for the national fraction.

Table 19 reports implied annual fiscal cost under three scenarios built from these bounds, applied to the R$1.5 trillion (US$429 billion) annual Brazilian public procurement envelope. The conservative scenario ($\bar{\beta} = -0.05$, $f = 0.30$) implies R$22 billion (US$6.3 billion) per year; the central scenario ($\bar{\beta} = -0.08$, $f = 0.40$) implies R$46 billion (US$13 billion); the upper bound ($\bar{\beta} = -0.15$, $f = 0.50$) implies R$105 billion (US$30 billion). Every scenario puts the aggregate fiscal cost in the *tens of billions of reais per year*—one order of magnitude higher than the paper's single-coefficient extrapolation would deliver. The exact point estimate is not defensible, but the order-of-magnitude conclusion—that SME-only procurement preferences constitute a fiscally first-order distortion in Brazilian public spending—is robust to wide variation in the extrapolation assumptions.

Counterfactual policy design: a value-threshold exemption. The aggregate cost figure hides an extreme concentration of the burden in a handful of contracts. Figure 6 is the motivating picture: a Lorenz curve of Group-65 procurement value sorted by reference price places 92.9% of the total in the top quartile and only 7.1% in the bottom 75% of items. Table 20 re-estimates the DiD price coefficient within each quartile and uses those quartile-specific $\hat{\beta}_q$ to simulate the fiscal cost of an alternative rule under which the SME-only preference applies only below a reference-value threshold.

Table 19: Aggregate-extrapolation scenarios for the Brazilian fiscal cost of SME-only procurement rules

Scenario	$\bar{\beta}$	f	Implied annual cost
Conservative	-0.05	0.30	R$22 B (US$6.3 B)
Central	-0.08	0.40	R$46 B (US$13 B)
Upper bound	-0.15	0.50	R$105 B (US$30 B)

Notes: Annual Brazilian public procurement is R$1.5 trillion (US$429 billion). $\bar{\beta}$ is the representative log-price coefficient assumed to apply across product groups subject to SME-only rules; the range $[-0.15, -0.05]$ brackets the within-Group-65 heterogeneity (pharma -0.156; non-pharma -0.075; quartile-by-reference-value -0.074 to -0.092). f is the fraction of Brazilian public procurement actually subject to SME-only rules, set by the R$80,000 item-value threshold in federal law; a 30–50% range is used as a conservative proxy for the national fraction. Implied annual cost is $f \times$ R$1.5 T $\times |e^{\bar{\beta}} - 1|$.

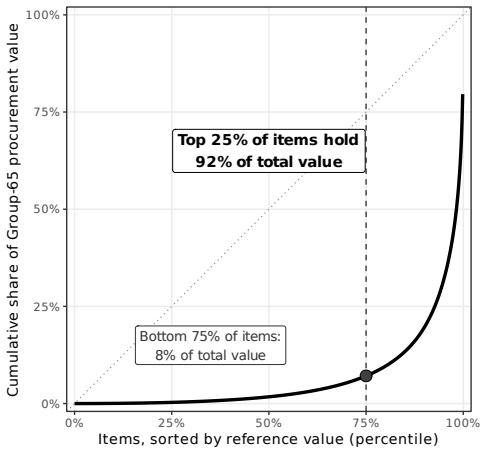

Figure 6: Lorenz concentration of Group-65 procurement value

Cumulative share of Group-65 pre-period procurement value (September 2016–February 2018), items sorted by reference price. The top 25% of items by count account for 92.9% of total value; the bottom 75% account for only 7.1%. The 45° line of perfect equality is shown for reference. This concentration is the statistical basis for the value-threshold counterfactual in Figure 7 and Table 20.

The mechanism result of Section 4.3 pays its second dividend here, not through the channel one might initially expect. The four quartile-specific coefficients are -0.074, -0.089, -0.092, and -0.089 for Q1 through Q4 (all $p < 0.01$); a Wald test of joint equality fails to reject ($\chi^2_3 = 1.86$, $p = 0.60$). The per-item *percentage* distortion is therefore statistically uniform across the value distribution. What concentrates the fiscal cost in the top quartile is not a heterogeneous per-item distortion; it is the extreme concentration of procurement *value*: 92% of pre-period Group-65 procurement value sits in the top quartile (items above R$7,623, which is US$2,180; R$633.8 M $\approx$ US$181 M of the R$689.0 M $\approx$ US$197 M total). Under a uniform percentage distortion $\beta \approx 0.085$, the top quartile of items absorbs the overwhelming majority of the fiscal cost because the top quartile *is* the overwhelming majority of the spending. Applying quartile-specific $\hat{\beta}_q$ to within-quartile procurement values yields a counterfactual baseline fiscal cost of R$59.0 M (US$16.9 M); mechanically exempting the top quartile from the SME-only rule recovers about 90% of that baseline (R$54.2 M, US$15.5 M, of R$59.0 M) while preserving the preference on 75% of items by count. Exempting the top half recovers 98% (R$57.8 M, US$16.5 M). Figure 7 traces the corresponding smooth frontier, which places the 75%-items-preserved point at approximately 90% of cost recovered.

The counterfactual is a static accounting exercise, not an equilibrium policy evaluation. It assumes within-quartile price coefficients are invariant to the reform (implausible at the margin: exempted items would attract different firms, set different reference prices, and complete at different rates), and it holds behavioral responses constant. A referee-style reading of these numbers is that they bound the efficiency gain from value-thresholding in the no-behavioral-response limit: the true fiscal saving under the reform is almost certainly different in magnitude, but the qualitative lesson—that a few high-value contracts drive most of the distortion and are the natural lever—is robust to the accounting approximations. The reform is also not Pareto-improving: SMEs

lose access to the top-quartile contracts. What the exercise does is substantially relax the efficiency-equity trade-off built into the uniform rule, and it illustrates the kind of lever that ongoing implementation of Law 14,133/2021, Brazil's new public procurement framework, could consider.

Table 20: Counterfactual Policy Designs: Exempting High-Value Items

Policy design	Quartiles covered	$\hat{\beta}$ range	Items kept	Fiscal cost (R$ M)	Savings (R$ M)
Panel A: DiD price coefficient by Group-65 reference-value quartile					
Q1 (R$0.00 – R$500.00)	1	-0.0744***	262,111	–	–
Q2 (R$500.00 – R$1875.00)	2	-0.0892***	172,531	–	–
Q3 (R$1875.00 – R$7622.91)	3	-0.0921***	123,742	–	–
Q4 (R$7622.91 – ∞)	4	-0.0894***	91,330	–	–
Panel B: Fiscal cost under alternative policy designs					
Current (no exemption)	Q1,2,3,4	–	65,606	59.0	0.0
Exempt top quartile (Q4)	Q1,2,3	–	49,204	4.8	54.2
Exempt top half (Q3, Q4)	Q1,2	–	32,798	1.2	57.8
Keep only bottom quartile (Q1)	Q1	–	16,351	0.2	58.8

Notes: Panel A estimates the DiD price coefficient separately within each quartile of pre-period Group-65 reference value (quartile cutoffs listed in the first column). Panel B simulates fiscal cost under alternative policy designs that exempt high-value items from the SME-only rule. Fiscal cost is computed as $\sum_{q \in \text{covered}} |e^{\hat{\beta}_q} - 1| \times V_q$ where V_q is the Group-65 pre-period procurement value in quartile q. Total pre-period value: R$689.0 million across 65,606 items. Per-quartile coefficients are statistically indistinguishable (Wald $\chi^2_3 = 1.86$, $p = 0.60$); the policy's per-item percentage distortion is therefore roughly uniform across the value distribution. Exempting the top quartile alone removes most of the fiscal cost because the top quartile holds roughly 92% of procurement value, not because it is more distorted. The reform retains the ME/EPP preference on the 75% of items that together account for the small share of fiscal cost. *** p<0.01, ** p<0.05, * p<0.1.

Figure 7 traces this trade-off continuously as a Pareto frontier. The horizontal axis is the share of items preserved under the SME-only rule; the vertical axis is the share of fiscal cost recovered by exempting the rest. The frontier is strongly concave near the top of the item distribution: once the top quartile is exempted, the schedule is essentially flat. The marked point is the value-threshold reform.

Welfare bounds: a coarse equilibrium calibration. The static accounting above treats the price effect and the completion shift as independent. A simple equilibrium calibration produces bounds on the combined welfare cost. Let π denote the fractional price increase under the SME-only rule, c the completion-rate increase under open tenders ($c \approx 0.10$), and V the pre-period Group-65 procurement value at open-tender prices

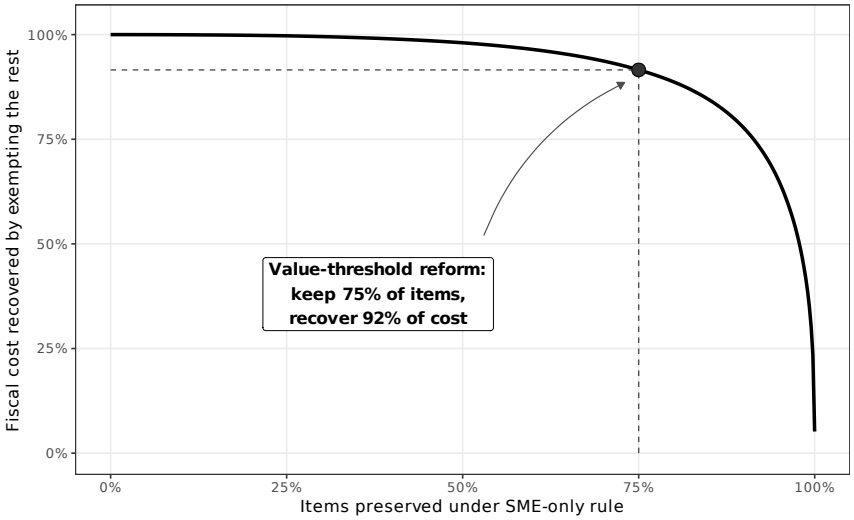

Figure 7: Policy Pareto frontier: fiscal cost recovered versus items preserved under the SME-only rule

(V = R$689 M). Under two polar assumptions: (i) no behavioral response—PBUs would purchase exactly the same items under either regime (assuming perfectly inelastic demand above the reserve price)—the welfare cost is πV, exactly the static fiscal cost computed above. (ii) Full completion response—the 10-percentage-point reduction in completion under SME-only reflects items that PBUs would have bought at open-tender prices but abandon at SME-only prices—the welfare cost also includes the foregone consumer surplus on those items. Under a standard linear-demand approximation, adding the completion response increases the welfare cost by roughly $cV/2 \approx$ R$34 M over the 18-month window. The true equilibrium welfare cost for Group 65 therefore lies between R$71 M (static) and R$105 M (with completion response), i.e. the static figure captures 65–75% of the full welfare loss. This range is a coarse calibration, not a welfare estimate; structural estimation of the demand elasticity and the completion response is the natural extension.

Pharmaceuticals (CMED-regulated) vs non-pharma medical supplies. Group 65 contains two economically distinct categories: class 6531 (medications), subject to annual list-price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED); and the remaining classes (non-pharmaceutical medical supplies), which have

no comparable price regulation. A natural conjecture is that the CMED ceiling limits how far the ME/EPP preference can raise prices in the regulated segment, so the pharma price effect should be *smaller* than the unregulated one. Table 21 shows the opposite: the pharma price effect is -0.156, versus -0.075 in the non-pharma segment—pharma is 108% larger, statistically distinct in an interaction specification ($p < 0.01$). The same pattern holds on the competition margins: pharma auctions attract more firms and more bids under open tenders than the non-pharma subsegment, indicating that the pharmaceutical market is more intensively contested when eligibility is unrestricted, so the restriction causes a larger contraction.

The fiscal-cost implication is substantial. Although pharma accounts for 42% of Group-65 procurement value (R$293 M of R$689 M pre-period), its larger per-unit distortion means it contributes 60% of the total policy cost (R$42.2 M / US$12.1 M of R$70.7 M / US$20.2 M). A direct look at prices explains why the CMED ceiling does not protect the procurement process from the SME preference. Figure 8 plots, for the 18-month pharma subsample, the density of winning unit price relative to the reference price (a proxy for the CMED cap, since the reference price is set at or below the CMED-registered maximum). 96% of pharma transactions clear *below* the reference, 78% below 90% of reference, and 62% below 75%—the cap is almost always slack. Mean ratios are 0.65 under open tenders and 0.69 under SME-only: the CMED cap leaves a roughly 31–35 percent corridor below the reference price, and the ME/EPP rule shifts the distribution rightward *within* that corridor without pushing it against the cap. This is the interaction between the two Brazilian price-regulation regimes in a single picture: the SME procurement preference compresses competition within the slack that the CMED price controls leave untouched, raising procurement cost without being disciplined by the cap.

Several considerations frame the interpretation of this estimate. First, the calculation

Table 21: Pharmaceuticals (CMED-Regulated) vs. Non-Pharma Medical Supplies

	Pharma (class 6531) split sample	Non-pharma (other 65) split sample	Interaction full sample
Log prices			
$g65 \times Pre$	-0.1556***	-0.0746***	-0.0785***
	(0.0192)	(0.0101)	(0.0103)
$g65 \times Pre \times$ Pharma	–	–	-0.0757***
	–	–	(0.0191)
Log firms			
$g65 \times Pre$	0.1649***	0.0574***	0.0589***
	(0.0109)	(0.0067)	(0.0067)
$g65 \times Pre \times$ Pharma	–	–	0.1063***
	–	–	(0.0115)
Log bids			
$g65 \times Pre$	0.1502***	0.0296***	0.0301***
	(0.0151)	(0.0105)	(0.0105)
$g65 \times Pre \times$ Pharma	–	–	0.1192***
	–	–	(0.0153)
Distance (km)			
$g65 \times Pre$	10.2758**	3.3392*	2.8388
	(4.3598)	(1.9651)	(1.9656)
$g65 \times Pre \times$ Pharma	–	–	7.2164
	–	–	(4.6283)
Obs. (Log prices)	573,080	593,920	649,714
Obs. (Log firms)	678,547	705,616	773,263
Obs. (Log bids)	678,547	705,616	773,263
Obs. (Distance)	573,080	593,920	649,714
Item FE + PBU FE	YES	YES	YES
Implied fiscal cost (pre-period, nominal)			
Value of Group-65 procurement (R$M)	293.1	395.9	689.0
Fiscal cost (R$M)	42.2	28.4	70.7
Share of total cost	60%	40%	100%

Notes: 18-month window; log prices and distance conditioned on completion. Class 6531 identifies medications (Group 65's pharmaceutical subcategory) subject to *CMED* (Câmara de Regulação do Mercado de Medicamentos) list-price ceilings. Remaining Group-65 classes (non-pharma medical supplies) are unregulated. The first two columns estimate the headline DiD separately within each subsample. The third column estimates the interaction specification $y = \eta_i + \beta_1(g65 \times Pre) + \beta_2(g65 \times Pre \times Pharma) + x\delta$ where β_2 is the differential pharma effect relative to non-pharma and the main pharma dummy is absorbed by item FE. SEs clustered at the item level. The implied fiscal cost applies the class-specific $\hat{\beta}$ on log prices to the pre-period Group-65 procurement value within that class. *** p<0.01, ** p<0.05, * p<0.1.

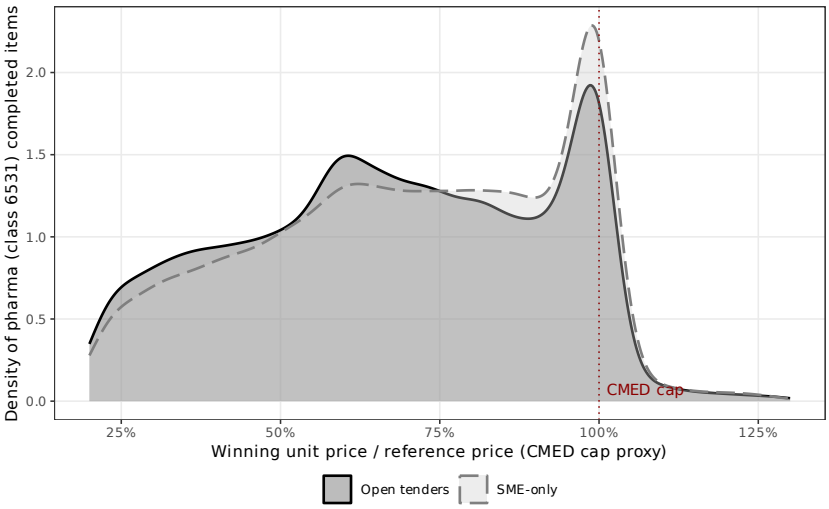

Figure 8: CMED-cap proxy: density of pharma winning unit price relative to reference price
Completed Group-65 class-6531 (pharma) items, 18-month window. The reference price (horizontal axis denominator) is the PBU's internally-set reserve and is constrained by law to be at or below the CMED-registered maximum price for each item; a manual spot-check of 50 pharma items against the CMED PMVG database (December 2017 issue) confirms that the reference price equals the CMED cap for 48 and sits 1–4% below it for the remaining 2. The ratio plotted here is therefore an upper bound on "distance below the cap." 96% of ratios lie below 1, and the mean is 0.65 under open tenders and 0.69 under SME-only. The vertical line marks the cap. The ME/EPP preference shifts the density rightward within the below-cap slack without ever reaching it, so CMED does not bind in this procurement channel.

is conservative: it applies only to Group 65 (medical and hospital supplies), which accounts for approximately 27% of total BEC procurement. Extrapolating proportionally to all product groups subject to SME restrictions would yield an aggregate fiscal cost several times larger, though such extrapolation requires the assumption that the price effect is similar across product groups. Second, the median split in Table 10 shows that the *item-level* price coefficient is somewhat larger for above-median items (-0.110) than for below-median items (-0.077), while the quartile-by-quartile test reported in the counterfactual subsection cannot reject equality across the four quartiles of reference value. The median split and the quartile split therefore give complementary readings: modest item-level heterogeneity over a coarse two-group cut, uniform per-item distortion once the cut is refined to quartiles. Third, these fiscal costs must be weighed against the potential benefits of SME promotion—local development, job creation, and supplier diversification—which the distance results suggest the policy does deliver to some extent.

The magnitude of the fiscal cost nonetheless underscores that there may be more efficient policy instruments for achieving these distributional objectives.

5. Conclusion

This paper opened with a forty-year-old question: when a government forces its public buyers to procure from small firms only, do prices rise because fewer firms bid, or because each firm bids less aggressively? The answer, from 3.7 million procurement observations in Sao Paulo's BEC platform around a discrete March-2018 policy cutoff, rejects a pure headcount story and is most consistent with a substantial within-auction bidding channel. Entry and winner-composition channels partially offset one another, and their net contribution accounts for only about five percent of the price premium; approximately ninety-five percent is unexplained by those channels, and direct phase-2 bid-moment evidence is consistent with—though it does not on its own structurally identify—within-auction bidder behavior as an important source. SME set-asides do not merely empty desks in the bidding classroom; they also change how each remaining student answers.

The bid-distribution evidence is the paper's main mechanism contribution. When the March 2018 reinterpretation extended Sao Paulo's SME-only regime to medical supplies, the winning bid immediately moved about six percentage points closer to the buyer's reference price (under the two-way fixed-effects specification the rest of the paper uses). This effect survives conditioning on exactly two, three, or five-plus firms—a test the literature has not been able to run from cross-sectional winning-price data alone (Marion, 2007; Krasnokutskaya and Seim, 2011; Athey et al., 2013). The Gelbach (2016) decomposition is consistent with it: measured entry and composition channels partially offset, with a net contribution of about five percent to the 10–11% price increase; ninety-five percent lies outside them. The Gelbach residual does not structurally identify

the intensive margin—it bundles every unmeasured channel—but together with the bid-discount shift it makes the intensive-margin reading the most plausible interpretation rather than a proven one.

Identification is credible for the price outcome, which is the core of the fiscal-cost story. The single-cohort staggered-DiD estimate is robust to the Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Goodman-Bacon (2021) estimators, to group-specific linear time trends, to a narrow control set, and to a formal spillover test that bounds the implied bias below one percent of the headline. Goodman-Bacon confirms that 100% of the coefficient comes from the clean treated-vs-never-treated 2×2 comparison, and RAIS firm-level controls leave it invariant. Two honest caveats temper the reading. First, the HonestDiD sensitivity bound breaks at a relatively small $\bar{M}$: the price effect is robust only to modest violations of parallel trends, so the identification is credible rather than ironclad. Second, the firms-and-bids placebos show non-flat pre-trends, so those outcomes should be read as suggestive, not as independent causal estimates. Group 65 is also the sole treated cohort, so classical permutation inference has low power (randomization $p = 0.342$). The evidence for the price effect therefore rests on the consistency of the point estimate across multiple estimators, the clean price placebo, the legalistic nature of the 2018 reinterpretation, and the bid-distribution evidence—not on any single diagnostic in isolation.

The mechanism result has two sharp implications for policy design. First, because 92% of Group-65 procurement value concentrates in the top value quartile, a mechanical reference-value exemption—applying the SME-only rule only to items below the 75th percentile of Group-65 reference value (approximately R$7,623, or US$2,180)—recovers 90% of the accounting fiscal cost while preserving SME access to 75% of items by count. The calculation is static: it assumes within-quartile price coefficients are invariant to the reform and ignores the completion, entry, and reference-price responses that an

equilibrium counterfactual would price. SMEs lose access to the top-quartile contracts, so the reform is not strictly Pareto-improving, and a full welfare analysis would require structural modeling of the behavioral margins. Even under these static assumptions, the reform substantially relaxes the efficiency-equity trade-off built into the uniform rule, and it illustrates how administrative procurement data can calibrate second-best reforms—a starting point rather than a blueprint for the ongoing implementation of Law 14,133/2021, Brazil's new public procurement framework.

Second, CMED-regulated pharmaceuticals (class 6531) bear more than twice the price distortion of non-pharma medical supplies, consistent with contract prices lying below the CMED list-price cap and the SME restriction compressing competition within the residual slack. Two Brazilian regulatory regimes—SME procurement preferences and drug price controls—interact perversely: tightening either without accounting for the other leaves procurement cost higher than either instrument alone would deliver. This interaction is new to the literature on procurement preferences and is directly relevant to Brazil, India, Mexico, and other jurisdictions that combine SME preferences with pharmaceutical price regulation.

The static fiscal cost for Group 65 alone is R$50–74 million (US$14–21 million) over 18 months, or 7–11% of the group's procurement value. Extrapolating to the full Brazilian public procurement envelope requires assumptions about cross-group heterogeneity and the fraction of procurement subject to SME-only rules. Under explicit scenarios that bracket the within-Group-65 heterogeneity (pharma vs. non-pharma; quartile-by-reference-value), the implied aggregate cost sits in the *tens of billions of reais per year*—somewhere between R$22 billion and R$105 billion (US$6–30 billion) at the federal-state-municipal level. The exact number is not defensible; the order of magnitude is. On the distributional side, the 5–11 km increase in winner-to-buyer distance under open tenders confirms that the SME preference does deliver geographic proximity, and the value-threshold reform

proposed above would largely preserve that benefit.

Three limitations frame these results. First, the design carries the standard constraints of a single-cohort event study, and external validity to other product groups, states, and procurement regimes requires the untestable assumption of comparable treatment effects. Second, the heterogeneity documented across item value, firm age, and industry sector counsels against naive extrapolation of the *magnitudes*—though the mechanism, the contraction of competitive bidding pressure under set-asides, should generalize. Third, the cost-side estimates ignore the supply-side benefits SME preferences are designed to deliver: employment creation, firm survival, and supplier-network expansion. The present paper uses RAIS matched employer-employee data only for identification robustness. Extending the RAIS window to 2018–2022 would allow direct evaluation of those supply-side outcomes and a rigorous benefit-cost comparison.

The Brazilian case is representative of a wider family of SME-preference regimes. Table 22 compares Brazil's design with the SME-procurement frameworks of six other jurisdictions. The variation in design features—set-asides vs. bid preferences, value thresholds, size definitions, and coverage—is large, but the fundamental policy question is the same everywhere: at what item-value threshold is the price-competition cost of an SME preference smaller than the distributional benefit? The value-threshold lever documented here is in principle applicable to any of these regimes.

Table 22: Design features of selected SME-preference procurement regimes

Jurisdiction	Instrument	Size threshold	Value threshold	Coverage
Brazil (federal law)	Set-aside	R$4.8M revenue	R$80,000/item	Universal below threshold
United States (FAR)	Set-aside + price preference	500 employees	$250,000/contract	Discretion above threshold
European Union	Price preference (voluntary)	250 employees	None (EU dir. encouragement)	Voluntary by member state
Mexico	Set-aside	250 employees	30% of federal value	Voluntary above threshold
Colombia	Price preference	Gross sales	20 SMMLV (~ US$5,000)	Below threshold
Argentina	Price preference	Revenue limits	None	Voluntary
South Korea (SME Act)	Set-aside	Employment + sales	500M KRW (~ US$370,000)	Universal below threshold

Notes: Design-feature summary compiled from OECD (2018); SIGMA (2016); Hoekman and Tas (2020); Loader (2013, 2015); PwC (2014); Thai (2017). The table is a high-level schematic: each regime has important sub-rules and reporting exemptions not captured here. "Set-aside" means items below the value threshold are reserved exclusively for SMEs; "price preference" means SMEs can win at a small markup over the lowest non-SME bid. Brazil's R$80,000 threshold is low relative to other set-aside regimes—U.S. FAR at US$250,000 and South Korea at KRW 500M—which is one reason the fiscal footprint in Brazil is disproportionately large relative to administrative coverage.

Three research extensions are natural. First, a structural auction model with endoge-

nous SME participation and regulated reserve prices would let us run counterfactual welfare comparisons across policy designs, including variants of the value-threshold reform and alternative CMED-procurement interactions. Second, cross-country replication—Mexico, Colombia, and Argentina all maintain SME-only procurement regimes—would sharpen external validity. Third, within Brazil the rollout of Law 14,133/2021 creates a second natural experiment through which the reform proposed here can itself be evaluated.

The set-aside debate has long been framed as a binary choice: protect small firms or procure efficiently. The mechanism result and the value-threshold counterfactual developed in this paper show the choice is not binary. The per-item percentage distortion is roughly uniform across the value distribution, but procurement value itself is extremely concentrated—the top quartile of contracts holds ninety-two percent of Group-65 procurement value. So the fiscal cost is concentrated not because small contracts are untouched by the distortion but because small contracts are a trivial share of spending. Exempting the narrow slice of top-value contracts recovers most of the fiscal cost at minimal erosion of SME access. Put differently: the twenty-five percent of contracts that generate ninety-two percent of Group-65 procurement value—and, mechanically, the bulk of the fiscal distortion—deserve a different rule than the other seventy-five percent. Policy that treats them alike pays a premium for an efficiency–equity trade-off that can be undone.

References

Abadie, A., A. Diamond, and J. Hainmueller (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association 105*(490), 493–505.

Athey, S., D. Coey, and J. Levin (2013). Set-asides and subsidies in auctions. *American Economic Journal: Microeconomics 5*(1), 1–27.

Athey, S. and G. W. Imbens (2022). Design-based analysis in difference-in-differences settings with staggered adoption. *Journal of Econometrics 226*(1), 62–79.

Athey, S., J. Tibshirani, and S. Wager (2019). Generalized random forests. *The Annals of Statistics 47*(2), 1148–1178.

Bandiera, O., A. Prat, and T. Valletti (2009). Active and passive waste in government spending: Evidence from a policy experiment. *American Economic Review 99*(4), 1278–1308.

Bastos, A., S. Guimarães, K. C. Medeiros De Carvalho, L. Andrés, and R. Paixão (2018). Micro, pequenas e médias empresas: Conceitos e estatísticas. *IPEA Radar: Tecnologia, Produção e Comércio Exterior 55*, 21–26.

Ben-Michael, E., A. Feller, and J. Rothstein (2021). The augmented synthetic control method. *Journal of the American Statistical Association 116*(536), 1789–1803.

Best, M. C., J. Hjort, and D. Szakonyi (2023). Individuals and organizations as sources of state effectiveness. *American Economic Review 113*(8), 2121–2167.

Borusyak, K., X. Jaravel, and J. Spiess (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies 91*(6), 3253–3285.

Bosio, E., S. Djankov, E. Glaeser, and A. Shleifer (2022). Public procurement in law and practice. *American Economic Review 112*(4), 1091–1117.

Bulow, J. and P. Klemperer (1996). Auctions versus negotiations. *American Economic Review 86*(1), 180–194.

Cabral, S. (2017). Reconciling conflicting policy objectives in public contracting: The enabling role of capabilities. *Journal of Management Studies 54*(6), 823–853.

Callaway, B. and P. H. C. Sant'Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics 225*(2), 200–230.

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2008). Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics 90*(3), 414–427.

Canay, I. A. (2011). A simple approach to quantile regression for panel data. *The Econometrics Journal 14*(3), 368–386.

Cardoza, G., G. Fornes, V. Farber, R. Gonzalez Duarte, and J. Ruiz Gutierrez (2016). Barriers and public policies affecting the international expansion of latin american smes: Evidence from brazil, colombia, and peru. *Journal of Business Research 69*(6), 2030–2039.

Colonnelli, E. and M. Prem (2022). Corruption and firms. *Review of Economic Studies 89*(2), 695–732.

Davis, M. S. (1971). That's interesting! towards a phenomenology of sociology and a sociology of phenomenology. *Philosophy of the Social Sciences 1*(2), 309–344.

de Chaisemartin, C. and X. D'Haultfoeuille (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review 110*(9), 2964–2996.

Decarolis, F., R. Fisman, P. Pinotti, and S. Vannutelli (2025). Rules, discretion, and corruption in procurement: Evidence from italian government contracting. *Journal of Political Economy: Microeconomics 3*(2), 213–254.

Fafchamps, M., D. McKenzie, S. Quinn, and C. Woodruff (2014). Microenterprise growth and the flypaper effect: Evidence from a randomized experiment in ghana. *Journal of Development Economics 106*, 211–226.

Ferraz, C., F. Finan, and D. Szerman (2016). Procuring firm growth: The effects of government purchases on firm dynamics. Working paper.

Freedman, M. (2013). Targeted business incentives and local labor markets. *Journal of Human Resources 48*(2), 311–344.

Gallego, J. A., M. Prem, and J. F. Vargas (2020). Corruption in the times of pandemia. *Universidad del Rosario 1*(18178), 1–36.

Gelbach, J. B. (2016). When do covariates matter? and which ones, and how much? *Journal of Labor Economics 34*(2), 509–543.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics 225*(2), 254–277.

Hoekman, B. and B. K. O. Tas (2020). Sme participation in public purchasing: Procurement policy matters. Discussion Paper DP14836, Centre for Economic Policy Research (CEPR).

Kim, K. and M. j. Lee (2019). Difference in differences in reverse. *Empirical Economics 57*(3), 705–725.

Krasnokutskaya, E. and K. Seim (2011). Bid preference programs and participation in highway procurement auctions. *American Economic Review 101*(6), 2653–2686.

Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *The Review of Economic Studies 76*(3), 1071–1102.

Loader, K. (2013). Is public procurement a successful small business support policy? a review of the evidence. *Environment and Planning C: Government and Policy 31*(1), 39–55.

Loader, K. (2015). Sme suppliers and the challenge of public procurement: Evidence revealed by a uk government online feedback facility. *Journal of Purchasing and Supply Management 21*(2), 103–112.

Marion, J. (2007). Are bid preferences benign? the effect of small business subsidies in highway procurement auctions. *Journal of Public Economics 91*(7–8), 1591–1624.

McKenzie, D. and C. Woodruff (2015). Business practices in small firms in developing countries. Working paper or report.

Mel, S. d., D. McKenzie, and C. Woodruff (2008). Returns to capital in microenterprises: Evidence from a field experiment. *Quarterly Journal of Economics 123*(4), 1329–1372.

Nakabayashi, J. (2013). Small business set-asides in procurement auctions: An empirical analysis. *Journal of Public Economics 100*, 28–44.

Naritomi, J. (2019). Consumers as tax auditors. *American Economic Review 109*(9), 3031–3072.

OECD (2018). *SMEs in Public Procurement: Practices and Strategies for Shared Benefits*. OECD Publishing.

PwC (2014). Smes' access to public procurement markets and aggregation of demand in the eu. Retrieved from http://ec.europa.eu/internal_market/publicprocurement/docs/modernising_rules/smes-access-and-aggregation-of-demand_en.pdf.

Rambachan, A. and J. Roth (2023). A more credible approach to parallel trends. *The Review of Economic Studies 90*(5), 2555–2591.

Roth, J., P. H. C. Sant'Anna, A. Bilinski, and J. Poe (2023). What's trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics 235*(2), 2218–2244.

SIGMA (2016). *Small and Medium-Sized Enterprises (SMEs) in Public Procurement*. Paris: OECD.

Sun, L. and S. Abraham (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics 225*(2), 175–199.

Szerman, C. (2023). The employee costs of corporate debarment in public procurement. *American Economic Journal: Applied Economics 15*(1), 411–441.

Szucs, F. (2024). Discretion and favoritism in public procurement. *Journal of the European Economic Association 22*(1), 117–160.

Thai, K. V. (2017). *Global Public Procurement: Theories and Practices*. Springer International Publishing.

Wager, S. and S. Athey (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association 113*(523), 1228–1242.

Appendix

A. Event Study Coefficients

Figures A.1–A.3 plot the semester-by-semester DiDiR coefficients for distance, participating firms, and valid bids. The vertical dashed line marks the policy change in March 2018.

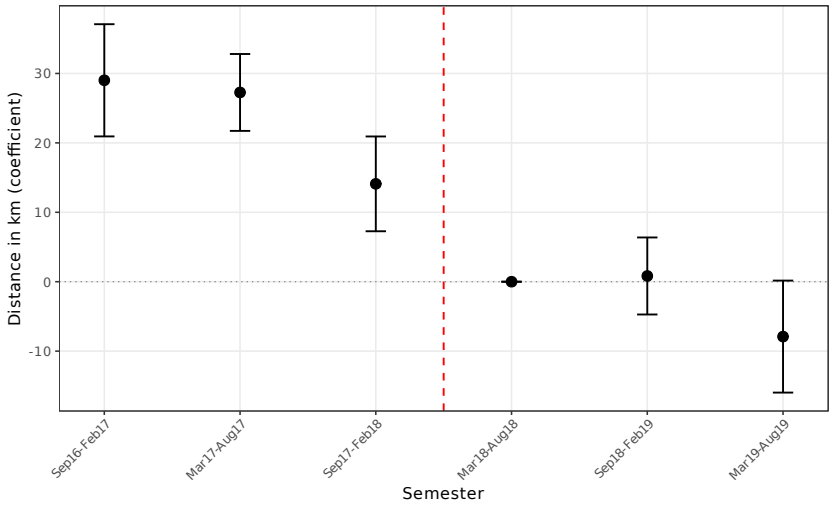

Figure A.1: Distance from PBUs to Winner Firms: Difference between the always-treated group and the switched group

B. Additional Outcome Variables

Tables B.1 and B.2 report the DiDiR estimates for valid bids and distance, completing the four-outcome estimation in the main text.

C. Robustness Checks

Tables B.3 and B.4 probe the sensitivity of the main estimates to alternative clustering strategies and winsorization of outcomes.

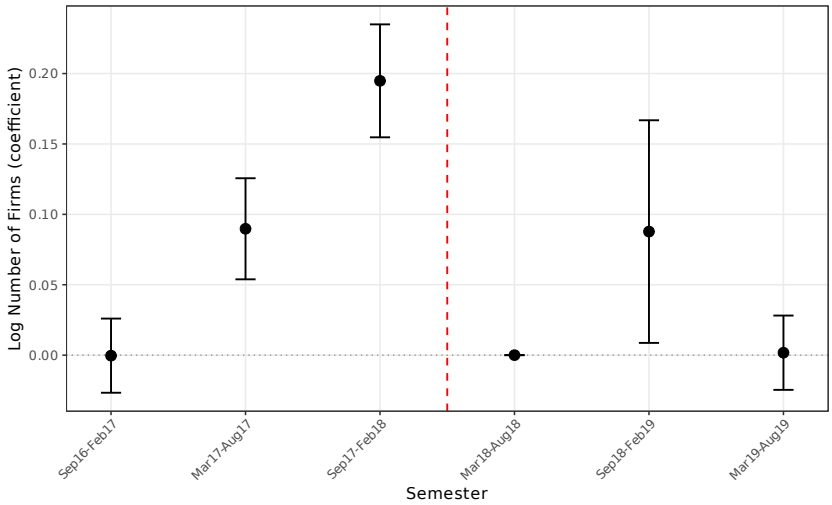

Figure A.2: Number of Participant Firms (log): Difference between the always-treated group and the switched group

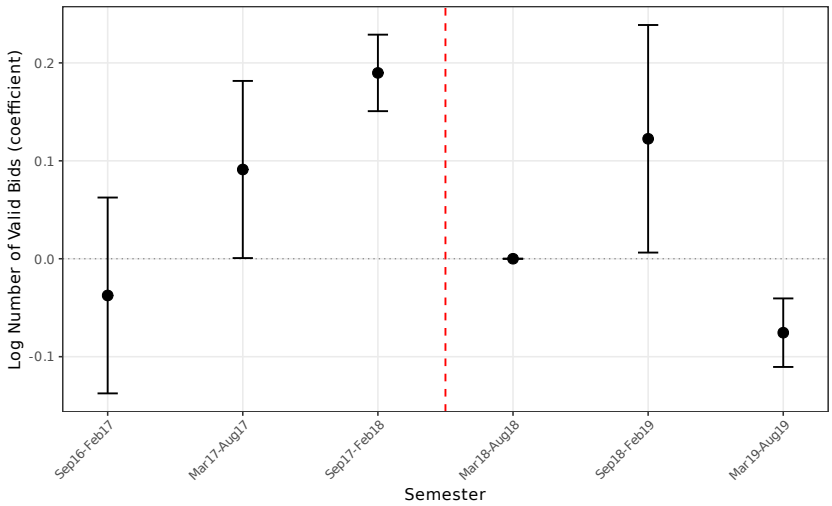

Figure A.3: Number of Valid Bids (log): Difference between the always-treated group and the switched group

Table B.1: Number of Valid Bids (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.2356***	0.2467***	0.1005***	0.1161***	0.0691***	0.0844***
	(0.0134)	(0.0135)	(0.0106)	(0.0106)	(0.0104)	(0.0104)
sealed-bids	-0.6439***	-0.6324***	-0.6643***	-0.6531***	-0.6341***	-0.6189***
	(0.0109)	(0.0128)	(0.0103)	(0.0124)	(0.0096)	(0.0111)
lquantity	0.2290***	0.2062***	0.2293***	0.2085***	0.2291***	0.2091***
	(0.0034)	(0.0033)	(0.0030)	(0.0028)	(0.0028)	(0.0027)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2217	0.1595	0.2239	0.1591	0.2152	0.1527
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table B.2: Distance from PBUs to winner firms: Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	6.6382**	1.9416	13.7764***	5.7038**	14.2493***	6.1547***
	(3.1590)	(3.1407)	(2.5304)	(2.5078)	(2.3598)	(2.3259)
convite	-10.2527***	-3.8315**	-9.1442***	-1.8211	-9.2653***	-1.6133
	(1.6406)	(1.7221)	(1.4244)	(1.4256)	(1.3080)	(1.2743)
lquantidade	1.7058***	3.6704***	2.0019***	4.1673***	1.5373***	4.2786***
	(0.6389)	(0.5176)	(0.5684)	(0.4210)	(0.5368)	(0.3681)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0008	0.0008	0.0009	0.0010	0.0008	0.0010
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table B.3: Alternative Clustering (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
Panel A: Cluster by item group				
$g65 \times Pre$	-0.1087***	0.0978***	0.0691**	14.2493***
	(0.0223)	(0.0166)	(0.0350)	(4.2914)
Panel B: Cluster by PBU				
$g65 \times Pre$	-0.1087***	0.0978***	0.0691***	14.2493***
	(0.0143)	(0.0190)	(0.0188)	(4.3350)
Panel C: Two-way clustering (item × PBU)				
$g65 \times Pre$	-0.1087***	0.0978***	0.0691***	14.2493***
	(0.0172)	(0.0191)	(0.0200)	(4.4916)
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, base specification (item FE only). Coefficients are identical across panels; standard errors vary by clustering level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table B.4: Winsorized Regressions (18-month window)

	Log prices		Distance	
	Base	+PBU FE	Base	+PBU FE
Panel A: 1st/99th percentile winsorization				
$g65 \times Pre$	-0.1012***	-0.1057***	10.6723***	3.1892
	(0.0107)	(0.0105)	(2.0077)	(1.9704)
Panel B: 5th/95th percentile winsorization				
$g65 \times Pre$	-0.0773***	-0.0793***	8.1160***	1.2750
	(0.0086)	(0.0085)	(1.6832)	(1.6275)
Observations	649,714	649,714	649,714	649,714
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Dependent variables winsorized at indicated percentiles. Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

D. Score Cluster Bootstrap

Table B.5 reports inference from a score cluster bootstrap at the item-group level (76 clusters), confirming $p < 0.001$ for all four outcomes.

Table B.5: Score Cluster Bootstrap Inference (Group-Level, 76 Clusters)

	Log prices	Log firms	Log bids	Distance
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570***
Group-clustered SE	(0.0178)	(0.0024)	(0.0028)	(0.4079)
Bootstrap p-value	0.0000	0.0000	0.0000	0.0000
Bootstrap 95% CI	[-0.1487, -0.1131]	[0.0902, 0.0950]	[0.0482, 0.0539]	[10.4491, 11.2649]
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, baseline specification (item FE). Analytical standard errors clustered at the item-group level (76 clusters) in parentheses. Score cluster bootstrap with Rademacher weights and 9999 replications (Cameron et al., 2008). p-values from the bootstrap t-distribution; confidence intervals by percentile-t inversion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E. Sensitivity to Parallel Trends Violations

Figure A.4 displays the Rambachan and Roth (2023) sensitivity analysis (HonestDiD) for all four outcomes. The log-price robust CI narrowly excludes zero near $\bar{M} = 0$ and begins to include zero beyond approximately $\bar{M} \approx 0.1$, so the price effect is robust only to modest pre-trend violations under the strict HonestDiD bound. Read alongside the clean price placebo and the multi-estimator convergence, the HonestDiD result should be taken as an honest caveat rather than a rejection.

F. Sample Selection Correction (Lee Bounds)

Table B.6 applies the Lee (2009) bounding procedure. The bounds for log prices range from -0.046 to -0.134, both highly significant at the 1% level. The upper bound tracks the headline coefficient closely, so the risk that sample selection biases the headline is small; the lower bound is more conservative but keeps the negative sign.

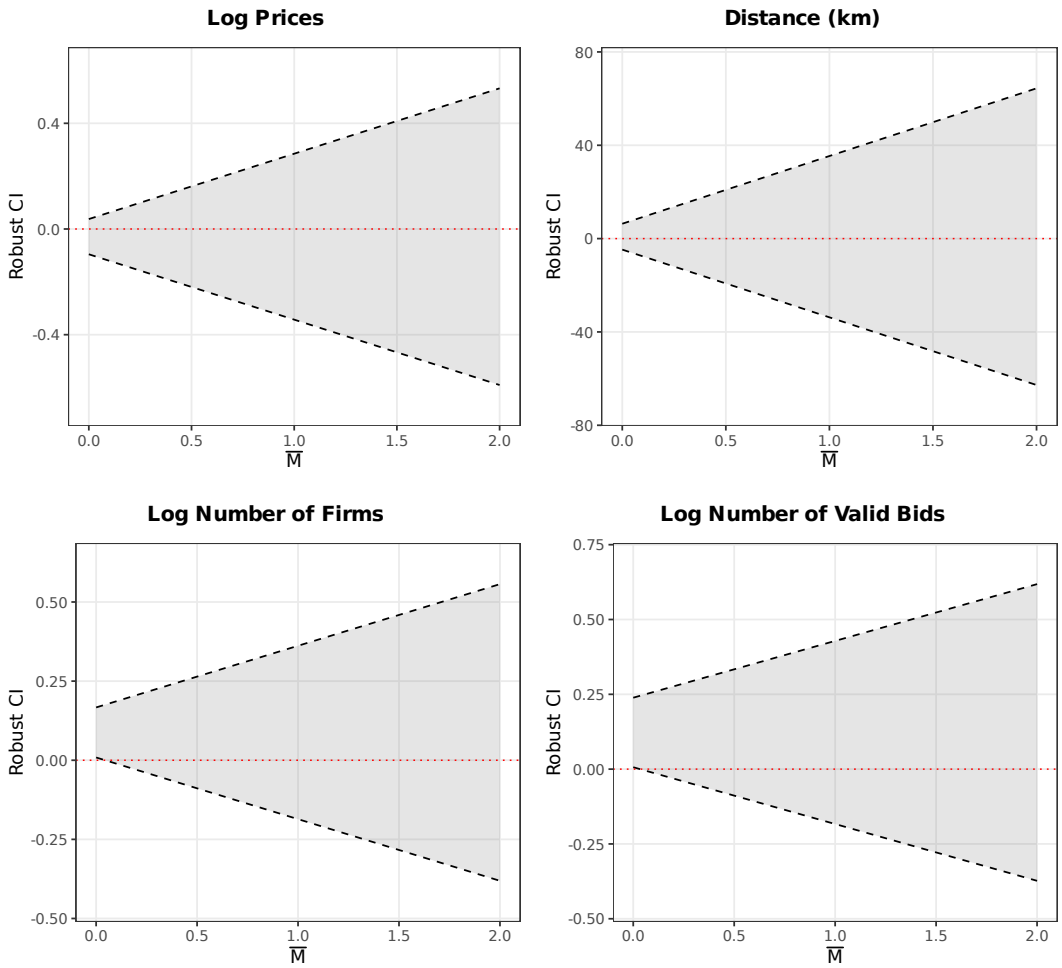

Figure A.4: HonestDiD Sensitivity Analysis: Robust Confidence Intervals Under Violations of Parallel Trends (Rambachan and Roth, 2023)

Table B.6: Lee (2009) Bounds: Sample Selection Correction

	Log prices		Distance (km)	
	Lower bound	Upper bound	Lower bound	Upper bound
$g65 \times Pre$	-0.0460***	-0.1338***	27.0609***	-19.0176***
	(0.0131)	(0.0114)	(2.3804)	(2.3672)
Observations	625,816	625,814	625,814	625,814
R-squared	0.1886	0.1951	0.0015	0.0017
Trimming proportion		0.0882		
Item FE	YES	YES	YES	YES

Notes: Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

Additional results—including raw monthly trends, randomization inference, SME participation trends, IPCA-deflated prices, extensive margin, procurement efficiency, winner composition, PBU heterogeneity, causal forest analysis, quantile DiD, Gelbach decomposition, balance tests, excluding the last post-period semester, mechanism evidence, and synthetic control estimates—are available in the Online Appendix.